Valuing Agglomeration*

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Abstract

State and local governments spend billions to attract corporate headquarters, promising localized economic spillovers. Using a novel dataset of U.S. headquarters relocations, I evaluate these claims by measuring how equity markets price incumbent firms when a peer's headquarters arrival is announced. Same-industry incumbents gain value, while incumbents in different industries do not show the same positive valuation response. The return premium decays with distance from the relocating firm and strengthens with product similarity. Adding a same-industry peer also increases local patenting. Therefore, relocation incentives are most defensible when they attract companies to locations with existing industry peers.

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1 Introduction

State and local governments spend billions of dollars annually on policies designed to attract corporate headquarters (Slattery and Zidar, 2020; Bartik, 2020). Governments offer these massive financial incentives expecting an incoming firm to boost local productivity for existing local firms. Existing evidence of spillovers captures realized outcomes such as jobs, productivity, and wages, but these outcomes take years to unfold and measure different dimensions of agglomeration. Understanding the value of colocation also requires weighing its anticipated benefits against costs such as rising rents and congestion. In this paper, I capture this expected net value for incumbent firms in a single forward-looking measure by examining daily stock market reactions to headquarters relocation announcements. This approach reveals which firms investors expect to benefit from gaining a nearby headquarters and how much those expected gains are worth to shareholders.

Relocations provide an ideal setting to study colocation because they change which firms are nearby. I examine changes in shareholder value for non-moving incumbent firms around 189 manually verified U.S. headquarters relocation announcements from 2006 to 2024. I find immediate positive repricing for nearby incumbents that gain a peer headquarters. When a firm announces a move, companies in the same industry near the new headquarters earn a 1.40% cumulative abnormal return over just two days. For the average exposed firm, this return implies an increase of about $70 million in market value around the announcement. I observe these positive returns for same-industry firms but not for different-industry firms. This pattern provides evidence of anticipated benefits from gaining an industry peer rather than the market pricing a broad regional signal. I document knowledge spillovers as a likely channel behind the same-industry return response because incumbent firms experience higher returns when they are geographically closer to the new headquarters and when they sell highly similar products. Finally, I show that patenting in the industry increases after the headquarters arrives. My findings suggest that relocation incentives are most defensible when they attract firms that deepen existing industry clusters.

What explains the positive repricing among same-industry incumbents? I examine three po-

tential mechanisms: labor market pooling, input sharing, and knowledge spillovers. Because labor market pooling and knowledge spillovers operate at different geographic scales, I first test the spatial reach of the returns. To measure this spatial decay, I compare the valuation response of same-ZIP incumbents to the response of incumbents within three and ten miles of the relocating firm. Labor markets span entire commuting zones (Rosenthal and Strange, 2020). Therefore, if labor pooling drives the effect, positive returns should extend out to the ten-mile radius. Conversely, the exchange of knowledge relies heavily on face-to-face interaction. These face-to-face benefits typically fade within a mile (Arzaghi and Henderson, 2008; Rosenthal and Strange, 2020; Baum-Snow et al., 2024). If knowledge spillovers explain the pattern, returns should concentrate within the relocating firm's ZIP code.

I find that the positive return is concentrated within the relocating firm's ZIP code. Although the hyper-local pattern is most consistent with knowledge flows, spatial proximity alone cannot completely isolate the mechanism. It leaves open the possibility of input sharing. Because corporate headquarters rely heavily on specialized local service inputs, they co-locate with dense concentrations of service providers (Davis and Henderson, 2008). Local supplier relationships therefore remain a plausible explanation for this spatial concentration. To separate these mechanisms, I test supplier connections and product similarity.

I test whether input sharing explains the return pattern by examining whether incumbent firms benefit when the arriving headquarters is a major supplier to their industry. Relocations primarily shift managerial and sales functions rather than shipping-intensive production (Strauss-Kahn and Vives, 2009). Consequently, these moves likely affect idea exchange more than the local flow of physical goods. Yet proximity still matters for service inputs requiring repeated face-to-face interaction. If proximity to input suppliers drives the valuation effect, downstream incumbents should benefit from the arrival. I find that incumbents connected to the relocating firm solely through a supplier relationship do not exhibit the positive valuation response seen in same-industry peers.

Next, I evaluate knowledge transfer by examining how relevant the new headquarters is to existing firms. Firms selling similar products should benefit more from knowledge transfer because the arriving firm's knowledge is more applicable to their own operations (Matray, 2021). Indeed, I find that the valuation response is stronger among firms with overlapping products, measured using the Text-Based Network Industry Classification of Hoberg and Phillips (2016). Together, the spatial concentration and stronger response among firms with similar products support the interpretation that investors anticipate localized knowledge spillovers.

To show that announcement returns reflect fundamental value rather than transitory sentiment, I examine whether realized outcomes evolve in the same direction as the initial market response. To examine patenting responses, I compare firms near the new headquarters with firms farther away in the same metropolitan statistical area and four-digit NAICS industry. These firms face similar regional conditions and industry trends but differ in their exposure to the headquarters arrival. Using a Poisson event-study model, I find that same-ZIP incumbents produce an average of 24 to 26 additional patents per year over the three years after the move, representing a 70% to 81% increase relative to the baseline predicted without an arrival. In contrast, neighboring incumbents in the same ZIP code that operate in different industries show no post-arrival increase in patenting, suggesting that neighborhood-wide growth alone cannot explain the results. This increase in realized innovation among same-industry peers supports the interpretation that the initial valuation gains reflect anticipated benefits from localized knowledge spillovers.

This paper contributes to the literature on place-based policies and local employment multipliers. While existing studies focus on job creation and employment spillovers (Slattery and Zidar, 2020; Bartik, 2020; Bartik and Sotherland, 2019; Gupta, 2020), my announcement-return analysis provides a single, forward-looking value. Because stock prices aggregate expectations about future cash flows, stock prices weigh the anticipated benefits of colocation against expected local costs like rising rents and congestion. By observing these returns for passive public incumbents, I can reveal which firms investors expect to benefit from a new local headquarters. I show that

headquarters attraction benefits shareholders primarily when it deepens an existing same-industry cluster rather than simply adding another large employer to a region.

I also contribute to the agglomeration literature by measuring the net present value of proximity. Prior work studies realized outcomes such as productivity, innovation, wages, employment, and housing prices (Glaeser et al., 1992; Greenstone et al., 2010; Moretti, 2021; Baum-Snow et al., 2024; Hu et al., 2024). I am the first to value agglomeration using daily stock returns. This approach offers three benefits. First, daily announcement windows capture the capitalization of expected spillovers before long-horizon outcomes become confounded by slow-moving local fundamentals. Second, my setting mitigates spatial sorting concerns. Productive firms sort into attractive regions, so it is hard to tell whether dense places generate spillovers or simply attract successful firms (Combes and Gobillon, 2015).[1] Because I observe passive incumbents, the headquarters arrival acts as a shock to the local cluster, complementing quasi-experimental studies that use large plant openings, bankruptcies, and regulatory shocks to isolate local spillovers (Greenstone et al., 2010; Bernstein et al., 2019; Matray, 2021).

Third, using incumbent returns allows me to separate valuation gains within the arriving firm's industry from gains across the broader local economy. Existing work attributes local growth either to industry diversity, with benefits spilling across sectors, or to specialization, with externalities confined to a single cluster (Rosenthal and Strange, 2004). Because I document positive repricing among same-industry incumbents alongside a near-zero response among different-industry incumbents in the same location, I provide evidence in favor of industry specialization over industry diversity.

Finally, I connect finance research on corporate geography to the broader literature on local firm spillovers. Related work studies how headquarters and plant proximity affects investment and productivity within the firm (Giroud, 2013). Corporate geography affects firm performance and shareholder value (Duchin and Sosyura, 2021; Dougal et al., 2022), while changes in local corpo-

[1] Dougal et al. (2022), for instance, show that young firms concentrate in high-valuation cities while mature firms relocate to lower-cost cities.

rate activity can affect the productivity and innovation of nearby firms (Greenstone et al., 2010; Matray, 2021). I bring these insights together by showing that investors capitalize a headquarters arrival into the values of neighboring firms, and I document which types of firms benefit. Thus I identify a financial-market dimension of local spillovers: corporate location decisions are reflected in the equity values of economically related firms outside the moving company.

The remainder of this paper is organized as follows. Section 2 describes the data construction and empirical strategy. Section 3 presents incumbent announcement returns. Section 4 examines labor pooling, input sharing, and knowledge transfers as possible mechanisms behind the return pattern. Section 5 reports realized changes in patenting following the relocation. Section 6 discusses alternative interpretations for my results, and Section 7 concludes.

2 Data and Methodology

2.1 Data Sources and Sample Construction

To build my sample of relocation events, I first identify headquarters (HQ) changes by comparing firm addresses across consecutive SEC 10-K filings from 2006 to 2024. To exclude administrative relabeling, I keep only persistent address changes where the new address appears in at least two consecutive filings. I then map ZIP codes to MSAs using the 2021 HUD ZIP-MSA crosswalk and assign each ZIP code to the MSA with the largest share of economic activity.

To validate candidate moves from 10-K filings, I use RavenPack News Analytics to compare the address in each article with the address reported in the filing. A valid relocation requires a headline identifying a primary corporate headquarters move that involves personnel movement, and the article text filters out incumbent firms related to mergers, financial headquarters moves, and partial relocations. The announcement date is the first public disclosure of the relocation, and I manually verify that the relocation is the primary disclosure in the news document. To

characterize these firms, I combine the relocation events with CRSP and Compustat for financial and stock return data, BEA input-output tables to construct sector-level supply-chain links and the Kogan, Papanikolaou, Seru, and Stoffman (KPSS) data to measure patenting (Kogan et al., 2017). Appendix Table A.1 in Appendix Section A provides the variable definitions.

Table 1 summarizes 189 verified headquarters relocation announcements spanning 16 distinct industry sectors. The destination-side and origin-side screens retain 185 and 177 announcements, respectively. The relocating firms are typically massive, driven by a few extremely large corporations that push the average market capitalization to $16.1 billion against a median of $1.5 billion. Manufacturing is the most common sector. Geographically, these relocations are dispersed across 51 destination and 69 origin metropolitan areas. The vast majority of these events represent major transitions, with nearly 78% involving moves across state lines.

Table 1. Headquarters Relocation Events

Notes: This table shows summary information for verified headquarters relocation announcements from 2006 to 2024. Market values of equity are fiscal-year-end values associated with the destination-headquarters observation and are reported in millions of dollars.

Panel A: Relocating Firms	
Relocation Events	189
Market Value of Equity, Mean	16,096
Market Value of Equity, Median	1,488
Unique Two-Digit NAICS Sectors	16
Most Common Sector	Manufacturing (NAICS 31–33)
Panel B: Relocation Geography	
Distinct Origin MSAs	69
Distinct Destination MSAs	51
Cross-State Relocations	147 (77.8%)
Top Destination MSAs	
1. Dallas–Fort Worth–Arlington, TX	25
2. Houston–Pasadena–The Woodlands, TX	14
3. Washington–Arlington–Alexandria, DC–VA–MD–WV	11
Top Origin MSAs	
1. New York–Newark–Jersey City, NY–NJ	24
2. Los Angeles–Long Beach–Anaheim, CA	18
3. San Francisco–Oakland–Fremont, CA	16

Figures 1 and 2 map the relocation events from Table 1. Arrivals are concentrated in the South, with Texas attracting the most relocating headquarters. Departures are concentrated in the West and Northeast, with California accounting for the most departures. At the MSA level, New York–Newark–Jersey City is the most frequent origin, while Dallas–Fort Worth–Arlington is the most frequent destination.

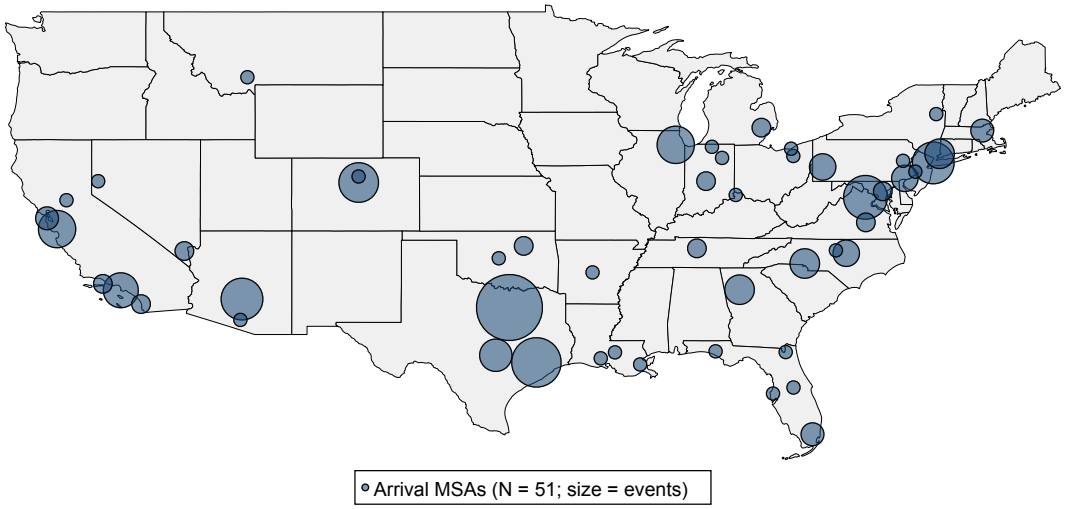


Figure 1. Headquarters Relocation Destinations

Notes: This figure shows the destination MSAs for verified headquarters relocation announcements from 2006 to 2024. Dot size is proportional to the number of arrivals in the MSA.

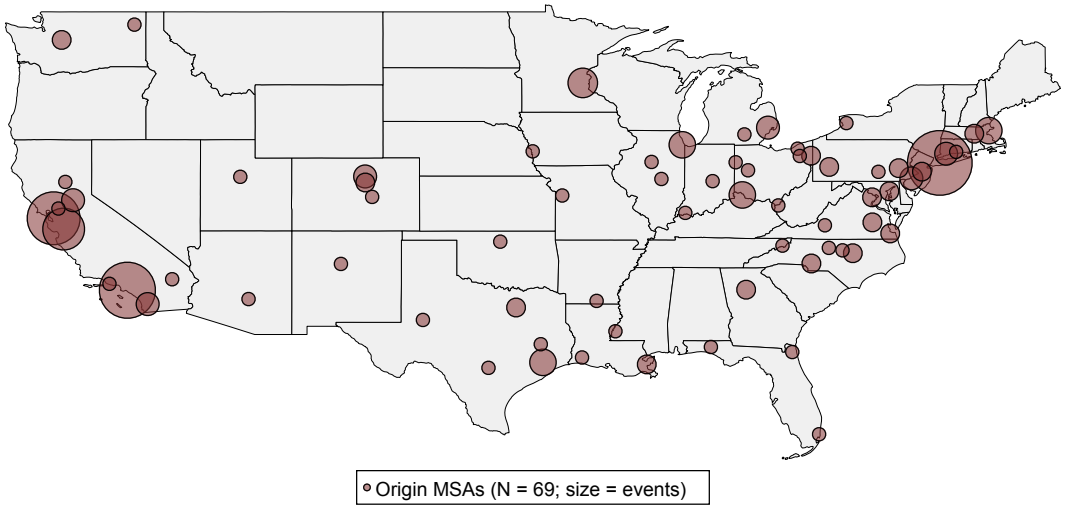


Figure 2. Headquarters Relocation Origins

Notes: This figure shows the origin MSAs for verified headquarters relocation announcements from 2006 to 2024. Dot size is proportional to the number of departures in the MSA.

A local incumbent is defined as a non-moving public firm headquartered in the relocating firm's relevant origin or destination ZIP code. Within this local set, I measure industry relatedness

by comparing the incumbent's NAICS code with the mover's code. My baseline same-industry definition requires a four-digit NAICS match. I then broaden the definition to three- and two-digit NAICS codes. At each NAICS level, different-industry incumbents share the mover's ZIP code but fall outside the displayed NAICS group. This structure allows me to test whether the valuation response is concentrated among narrowly defined industry peers or extends to a broader set of local firms. Appendix Table A.2 in Appendix Section B.1 reports the same- and different-industry sample sizes under each definition.

Table 2 details the four-digit same-industry public incumbents used in the headline return estimates. The final sample has 81 arriving firm-events and 31 departing firm-events. While these local peers are economically large, they are generally smaller than the relocating firms. Destination-side mean market equity approaches $5 billion, yet the median sits at just $785 million. Prior innovation is concentrated among a highly active subset of these firms, with roughly 40% holding at least one patent in the year before the relocation announcement.

Table 2. Summary Statistics for Local Incumbents

Notes: This table presents summary statistics for non-moving public incumbents in the relocating firm's ZIP code and four-digit NAICS industry. Market value and sales are reported in millions of dollars. Patent counts represent grants during the calendar year preceding the announcement.

	Mean	Median	St. Dev.
Panel A: Arrivals			
Incumbent firms	81		
Market Value of Equity	4,989	785	20,079
Sales	2,095	96	8,096
Patent Count	28	0	198
Panel B: Departures			
Incumbent firms	31		
Market Value of Equity	7,453	497	15,507
Sales	1,615	169	2,636
Patent Count	11	0	29

2.2 Identifying the Value of Agglomeration

I examine stock returns around the first public announcement of a headquarters relocation to measure how investors revise the value of non-moving incumbents in the origin and destination regions. Event day $t = 0$ is the first trading date on or after the public announcement date.[2] For each incumbent and announcement pair, I estimate expected returns using the 90 trading days from $t = -120$ through $t = -31$. This window provides a multi-month baseline and ends before the announcement to reduce contamination from information leakage or anticipatory trading. For each expected-return model, I require at least 80 days with nonmissing returns for both the incumbent and the relevant benchmark. I estimate the following market model by ordinary least squares:

$$R_{i,e,t} = \alpha_{i,e} + \beta_{i,e} R_{m,e,t} + \varepsilon_{i,e,t}, \tag{1}$$

where i identifies the incumbent, e identifies the relocation announcement, and t indexes trading days relative to that announcement. The dependent variable $R_{i,e,t}$ is the incumbent's daily stock return, and $R_{m,e,t}$ is the CRSP value-weighted market return, including dividends, on the corresponding calendar date. The intercept $\alpha_{i,e}$ is the incumbent's expected daily return when the market return is zero, and $\beta_{i,e}$ measures the incumbent's sensitivity to market movements. For example, consider Tesla's October 7, 2021 announcement that it would move its headquarters to Austin. Here, e identifies Tesla's announcement, and i identifies an eligible public incumbent already headquartered in the destination region. I estimate a separate market model for each eligible incumbent using its own pre-announcement returns and the corresponding market returns.

As an alternative benchmark, I estimate an industry model using the return on the incumbent's Fama–French 48-industry portfolio (Fama and French, 1997). I assign each incumbent to a portfolio using its SIC classification and estimate:

$$R_{i,e,t} = \alpha_{i,e}^{FF48} + \beta_{i,e}^{FF48} R_{j,e,t}^{FF48} + \varepsilon_{i,e,t}^{FF48}, \tag{2}$$

[2] Weekend and market-holiday announcements roll forward to the next trading date.

where j identifies the incumbent's Fama–French industry portfolio and $R_{j,e,t}^{FF48}$ is the industry return on the corresponding calendar date. The intercept $\alpha_{i,e}^{FF48}$ is the incumbent's expected daily return when its industry-portfolio return is zero, and $\beta_{i,e}^{FF48}$ measures the incumbent's sensitivity to industry returns. I estimate this model by ordinary least squares over the same pre-announcement window as the market model. For a semiconductor incumbent exposed to Tesla's arrival, for example, the benchmark is the Fama–French portfolio covering that incumbent's SIC classification.

Using the estimated coefficients, I calculate daily abnormal returns as the difference between observed returns and the returns predicted by the benchmark model. Under the market model:

$$AR_{i,e,t} = R_{i,e,t} - \left(\widehat{\alpha}_{i,e} + \widehat{\beta}_{i,e} R_{m,e,t}\right) \tag{3}$$

The predicted return combines the incumbent's pre-announcement relationship with the market and the market return on that trading day. I calculate industry-model abnormal returns analogously using the coefficients and industry return from equation (2). For example, if an incumbent exposed to Tesla's arrival earns 3% on a day when its market model from Equation 1 predicts 1%, its abnormal return is 2%. I repeat this calculation for each eligible incumbent on every trading day in the announcement window.

I then sum daily abnormal returns over the event window to obtain the cumulative abnormal return:

$$CAR_{i,e}(a,b) = \sum_{t=a}^{b} AR_{i,e,t}, \tag{4}$$

where a and b are the first and last included trading days. Each incumbent–announcement pair contributes one CAR for a given window and expected-return benchmark. The main results measure incumbent firms' cumulative abnormal returns over the announcement day and the following trading day. This short window allows prices to respond to announcements made after market close while limiting the influence of unrelated news. Continuing the Tesla example, if the incumbent's abnormal return is 2% on day $t = 0$ and 1% on day $t = 1$, its $CAR_{i,e}(0,1)$ is 3%.

For each exposure group (such as same- or different-industry firms), expected-return model, and event window, I pool eligible incumbent–announcement pairs across relocations and estimate the constant-only regression:

$$CAR_{i,e}(a,b) = \gamma_0 + \eta_{i,e}, \tag{5}$$

where γ_0 is the group's mean CAR and $\eta_{i,e}$ is each incumbent–announcement pair's deviation from that mean. For the same-industry arrival sample, the estimated intercept $\widehat{\gamma}_0$ is the average announcement-window CAR of incumbents in the arriving firm's industry and destination region. This sample includes eligible incumbents exposed to Tesla's headquarters arrival alongside those exposed to Boeing's arrival and other headquarters arrivals. I estimate arrivals and departures separately and cluster standard errors by relocation announcement to allow returns among incumbents exposed to the same move to be correlated.

I test $H_0 : \gamma_0 = 0$ to assess whether each group's mean CAR differs from zero relative to its expected-return benchmark. I repeat the analysis for same-ZIP incumbents that share the mover's four-, three-, or two-digit NAICS industry to examine how the estimated response changes as the industry definition broadens. At each NAICS level, I also estimate mean CARs for same-ZIP incumbents outside that industry definition as a comparison for broad local news.

2.3 Geographic Reach

To explore the mechanisms behind the return patterns, I next test whether the valuation gains are consistent with labor pooling, input sharing, or knowledge transfers (Rosenthal and Strange, 2004). I begin with geographic proximity because these channels operate over different distances (Rosenthal and Strange, 2020). Access to a shared labor market may extend across a commuting area, while knowledge exchange through frequent personal interaction may depend on closer proximity.

To measure proximity beyond the mover's ZIP code, I expand the sample to public incumbents in the mover's four-digit NAICS industry that are located within the same state and MSA but

are less than ten miles from the relocating headquarters. I use the destination headquarters for arrivals and the origin headquarters for departures. Distance is measured between the centroids of the incumbent's ZIP code and the mover's origin or destination ZIP code. Same-ZIP firms consequently have a measured distance of zero. Using these distances, I divide incumbents into three groups: firms in the mover's ZIP code, firms outside that ZIP code but less than three miles away, and firms located from three to under ten miles away. I estimate differences in mean CARs across these groups using the following specification:

$$CAR_{i,e} = \gamma_0 + \sum_{b=1}^{2} \gamma_b \text{DistanceBin}_{b,i,e} + \eta_{i,e}, \tag{6}$$

where b indexes the two distance groups outside the mover's ZIP code. The indicator $\text{DistanceBin}_{1,i,e}$ equals one when the incumbent is outside the mover's ZIP code but less than three miles away, and $\text{DistanceBin}_{2,i,e}$ equals one when it is located from three to under ten miles away. Same-ZIP firms form the omitted category such that γ_0 represents the mean CAR for very local firms. The coefficient γ_b measures the difference between distance group b and the same-ZIP group, making $\gamma_0 + \gamma_b$ the pooled mean CAR for that distance group.

Physical distance also depends on the ease of traveling between locations. Two headquarters can be close in straight-line distance but require a longer drive because a river, highway layout, or limited road access makes the route indirect. To account for this, I supplement geographic distance with estimated driving time using the same within 10-mile, same-industry sample as the distance sample. Using Open Source Routing, I obtain the duration of the fastest driving route from the incumbent's ZIP-code centroid to the mover's destination ZIP-code centroid. Using these distances, I construct the driving-time groups to distinguish trips of at most three minutes, three through nine minutes, and more than nine minutes. I then replace the distance indicators in equation (6) with indicators for the two longer driving-time groups, leaving the shortest-trip group of at most a three-minute driving commute as the omitted category. Each additional coefficient measures the difference for incumbents requiring a longer trip.

2.4 Product-Market Relatedness

While geographic proximity measures how easily firms can interact, product-market relatedness captures how closely their business activities overlap. To measure this product-market relatedness I use the Text-Based Network Industry Classification (TNIC) developed by Hoberg and Phillips (2016). TNIC compares the vocabulary firms use to describe their products and operations in their 10-K filings, allowing relatedness to vary across firm pairs and over time. I use TNIC-3, which identifies related firms at a level of industry detail comparable to three-digit SIC classifications, to form a sample of incumbent firms that have any positive textual link. Higher similarity scores indicate greater overlap in product-market vocabulary.

For example, two firms repeatedly discussing semiconductors and memory processors receive a higher score than a moving and incumbent firm pair whose descriptions center on unrelated goods. Specifically, Ring Energy and Earthstone Energy form a highly product-similar pair in my sample. TNIC identifies this relationship from the firms' overlapping 10-K product vocabulary, which includes words such as "oil," "gas," "exploration," "production," "drilling," and "wells."

To construct the local product-relatedness sample, I match arriving firms to TNIC peers headquartered in the destination ZIP code. I then use the similarity scores to form progressively more closely related groups of mover and incumbent pairs. Specifically, I calculate the 10th, 30th, and 50th percentiles of the score distribution. I apply equation (5) separately to these groups to estimate mean CARs at increasing levels of product similarity.

2.5 Supplier Links

Firms can also be connected through the inputs used in production. To capture these relationships, I construct an additional sample with supplier-linked incumbents using the BEA industry-by-industry data. I classify an incumbent as supplier-linked to a relocating firm when the mover's two-digit NAICS sector is the incumbent sector's largest upstream input source.

To illustrate this, consider Caterpillar's 2022 headquarters relocation to Irving, Texas. Caterpillar operates in the manufacturing sector, while Vistra is an electric utility already headquartered in the same ZIP code. The BEA input-output table indicates that the utility sector draws approximately 9.5% of its intermediate inputs from manufacturing throughout the supply chain, representing its largest external input requirement. Consequently, I classify Caterpillar as the supplier-side mover and Vistra as the exposed incumbent. The specific variable entering my test is Vistra's cumulative abnormal return around Caterpillar's relocation announcement. Ultimately, this setup asks whether Vistra's market value changes when a firm from its primary supply sector moves into its immediate neighborhood.

3 Valuing Agglomeration Spillovers

I begin by examining whether headquarters relocations increase or decrease the value of nearby firms in the same industry, using non-moving incumbents that share the relocating firm's ZIP code and four-digit NAICS industry. If proximity to an industry peer creates benefits that exceed the costs of greater competition for local resources, these incumbents should gain value when a headquarters arrival is announced and lose value when a departure is announced. Incumbent returns therefore measure how investors revalue nearby firms when an industry peer announces a headquarters move into or out of the local area.

Figures 3 and 4 report incumbent returns around headquarters arrival and departure announcements. The darker bars show returns for same-industry incumbents, and the lighter bars show returns for incumbents in other industries within the same ZIP code. The gray bars report the same-industry return minus the different-industry return, in percentage points, and their confidence intervals test whether the two groups' mean returns differ. Panel A uses the market model to estimate abnormal returns, and Panel B uses the Fama–French 48-industry model. The figures report cumulative abnormal returns over the announcement day and the next trading day, $[0, +1]$,

and over the three-day window that also includes the preceding trading day, $[-1, +1]$. Including the day after the news article (day 1) allows prices to respond to announcements made after the market closes, while adding the day before the announcement allows for information arriving shortly before the announcement.[3]

Figure 3 shows that same-industry incumbents gain value when a headquarters arrival is announced. Across 81 incumbent firm-events, the mean market-model return is 1.40% over $[0, +1]$ and 1.36% over $[-1, +1]$, significant at the 5% and 10% levels, respectively. The similarity of the estimates indicates that the positive response does not depend on excluding the trading day immediately before the announcement. The magnitudes are also economically meaningful. The average pre-announcement market value of these incumbents is $4.99 billion. Applying the two-day return to a firm of this size implies an increase in equity value of approximately $70 million.[4] Investors thus attach substantial value to news of an arriving industry peer.

One potential concern is that the positive returns associated with new firm arrivals reflect coincident national industry trends. To examine this possibility, I compare the market-model returns with the Fama–French 48-industry returns in Panel B of Figure 3. Both specifications show positive same-industry returns. Under the industry model in Panel B, the returns are slightly smaller in magnitude and they are still positive, with returns of around 1.06% in the two-day window and 1.16% when including the day prior to the announcement. This suggests that national industry trends alone cannot explain the positive return response for incumbent firms with a new headquarters arrival.

I next examine whether the gains extend to nearby firms in other industries. This comparison helps distinguish between two views of agglomeration. In the Jacobs view, urban diversity creates benefits as ideas combine across industries (Jacobs, 1969; Duranton and Puga, 2001). If these ben-

[3] Wider announcement windows preserve the positive arrival and negative departure estimates under both models. Appendix Figure A.1 reports windows extending up to five trading days on either side of disclosure. Confidence intervals generally widen as the windows expand.

[4] I multiply the mean two-day market-model return by the arrival-side mean market value in Table 2. This calculation illustrates the magnitude for a firm with the sample's mean market value. It does not estimate the average dollar gain across individual incumbents.

efits or increased demand from the arriving headquarters are broadly shared, nearby firms outside its industry should also gain value. In the Marshall–Arrow–Romer view, specialization creates externalities within local industry clusters (Marshall, 1890; Arrow, 1962; Romer, 1986; Glaeser et al., 1992; Henderson, 2003). The specialization view predicts a stronger response among firms that share the arriving headquarters' industry. The different-industry comparison also helps assess whether the announcement conveys favorable news about the destination's broader economy, which could raise the values of firms across local industries.

The lighter bars in Figure 3 show no corresponding appreciation among different-industry incumbents. Their market-model returns are actually slightly negative, reaching -0.33% over the two-day window and -0.51% over the three-day window. Across the two models and announcement windows, the estimates range from -0.51% to -0.19%, where the three-day market-model estimate is statistically significant at conventional levels. Comparing the same-industry returns with different-industry returns presented in this figure tells a similar story. The gray bars indicate that same-industry arrival returns exceed different-industry returns by 1.73 percentage points over $[0, +1]$ and 1.87 percentage points over $[-1, +1]$ under the market model. The corresponding differences under the industry model are 1.25 and 1.51 percentage points.[5] Thus, the evidence provides little support for a positive valuation effect shared broadly across nearby public firms.

For policymakers considering attraction incentives, this concentration of gains suggests that the value of spillovers depends on the arriving headquarters' relationship to existing local firms. The return pattern is more consistent with benefits from deepening an existing industry cluster than with gains distributed across the local economy. Therefore, a headquarters that fits the destination's industry base may create benefits for incumbent firms even when another headquarters of similar size would not.

Figure 4 examines the corresponding response when a headquarters announces that it will leave a region. Across 31 same-industry incumbent firm-events at the origin, mean market-model returns

[5]For a detailed comparison of differences across event windows, see Appendix Table A.3.

are -1.15% over $[0,+1]$ and -1.86% over $[-1,+1]$. The industry-model estimates are similar, showing -1.39% over the two-day window and -1.58% over the three-day window. The gray bars reinforce that the loss pattern is concentrated among same-industry peers that lost a headquarters from their ZIP code. The same-industry departure returns fall below different-industry returns by 1.20 percentage points over $[0,+1]$ and 1.95 percentage points over $[-1,+1]$ under the market model. The corresponding gaps under the industry model are 1.38 and 1.64 percentage points. The smaller sample of same-industry incumbents at origin locations contributes to the lower precision of the departure estimates. Nevertheless, the negative same-industry returns, together with near-zero returns in other industries, are consistent with investors expecting a loss of local benefits when an industry peer departs. Taken together, the arrival gains and departure losses are consistent with investors valuing proximity to an industry peer.

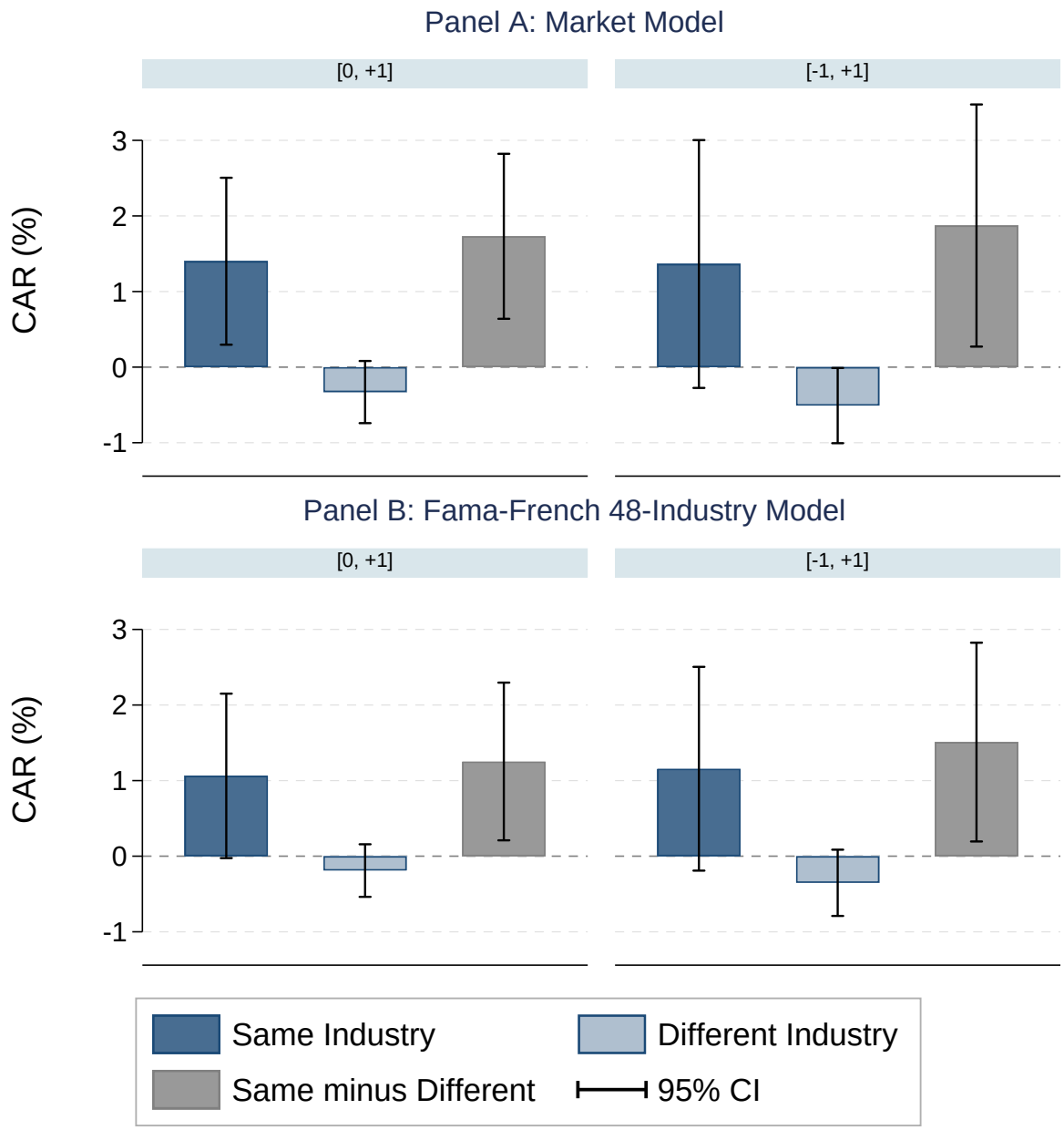


Figure 3. Incumbent Returns Around Headquarters Arrivals

Notes: This figure shows mean cumulative abnormal returns for destination-side public incumbents in the arriving headquarters' ZIP code. Same-industry firms share its four-digit NAICS industry. Different-industry firms do not. Gray bars show the same-industry mean minus the different-industry mean, in percentage points. Panel A uses the market model and Panel B the Fama–French 48-industry model. Each panel reports the $[0, +1]$ and $[-1, +1]$ windows, including both endpoints, with day 0 denoting the announcement. The same-industry sample contains 81 firm-events under both models. The different-industry samples contain 867 firm-events in Panel A and 858 in Panel B. Whiskers represent 95% confidence intervals using standard errors clustered by relocation announcement.

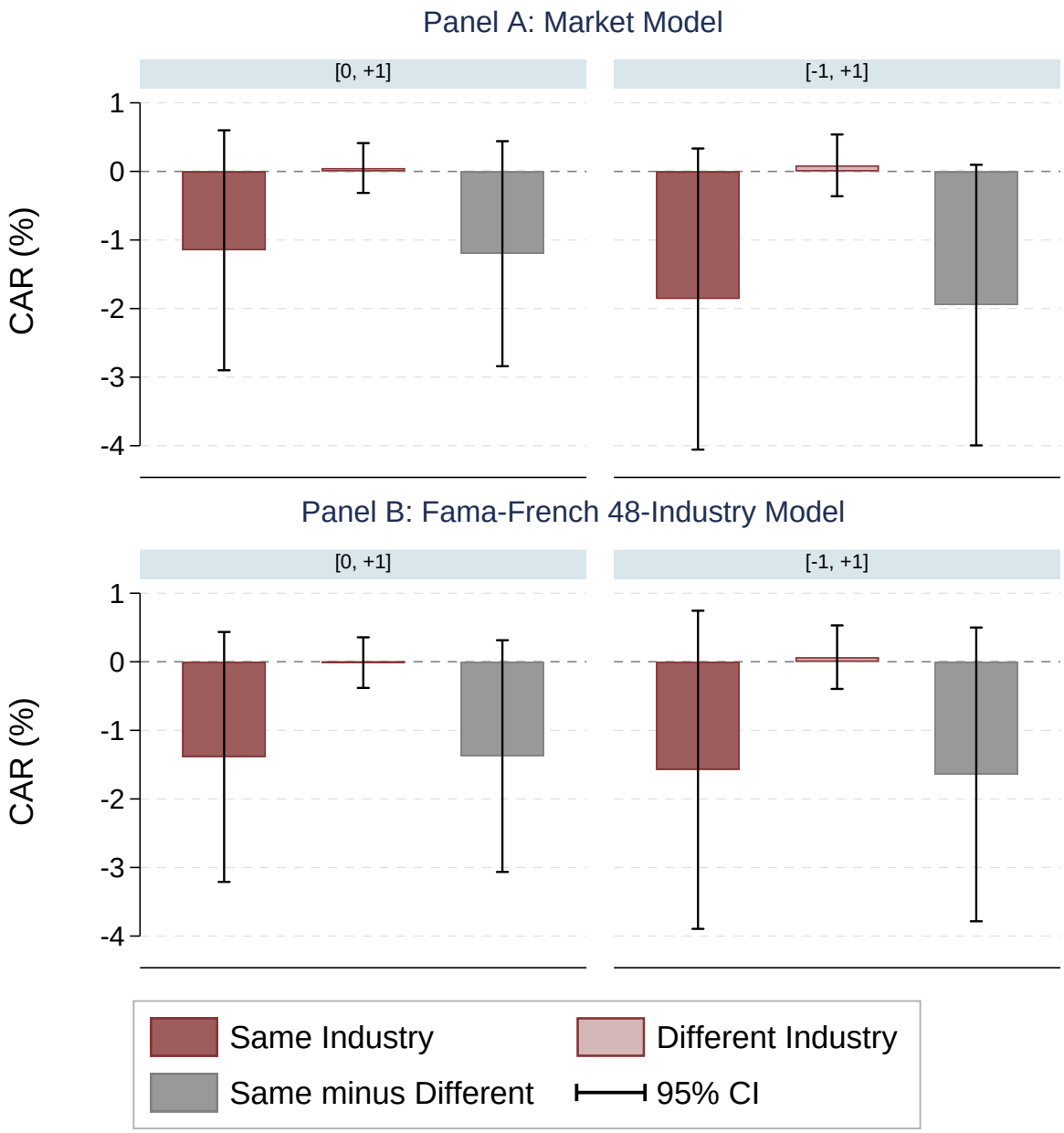


Figure 4. Incumbent Returns Around Headquarters Departures

Notes: This figure shows mean cumulative abnormal returns for origin-side public incumbents in the departing headquarters' ZIP code. Same-industry firms share its four-digit NAICS industry. Different-industry firms do not. Gray bars show the same-industry mean minus the different-industry mean, in percentage points. Panel A uses the market model and Panel B the Fama–French 48-industry model. Each panel reports the $[0, +1]$ and $[-1, +1]$ windows, including both endpoints, with day 0 denoting the announcement. The same-industry sample contains 31 firm-events under both models. The different-industry samples contain 728 firm-events in Panel A and 722 in Panel B. Whiskers represent 95% confidence intervals using standard errors clustered by relocation announcement.

Given that the announcement returns are similar under the market and industry models, the remaining tests use the market model. I next examine how the surrounding return paths relate to the two-day announcement response. If investors anticipate the move, some adjustment may precede the recorded announcement. Investors may also take time to assess the announcement, so prices can adjust both before and after disclosure. Disclosure can still convey positive or negative news by confirming the destination or resolving uncertainty about whether the move will occur. Announcement returns therefore depend on both the economic importance of the relocation and the information revealed at the announcement (Malatesta and Thompson, 1985).

Figure 5 plots cumulative abnormal returns from ten trading days before through ten trading days after the announcement. Panel A shows arrivals, and Panel B shows departures. Within each panel, same-industry incumbents appear on the left and different-industry incumbents on the right. Each path is normalized to zero on day -1, so the value on day 1 equals the two-day announcement return.[6] The left side of Panel A shows a positive and statistically significant two-day announcement response among same-industry incumbents. The gain does not immediately reverse, and cumulative returns generally rise further after disclosure. Different-industry incumbents on the right side of Panel A show no comparable appreciation. Thus, the concentration of gains among industry peers extends beyond the announcement window. Same-industry values also decline and recover before disclosure, although returns over the pre-announcement windows are imprecisely estimated. Panel A of Appendix Table A.3 examines arrival returns 20 days before and after the announcement and shows statistically insignificant pre-period returns and positive post-period returns.

The departure paths show the opposite pattern. Same-industry values on the left side of Panel B fall before and around the announcement, while different-industry values on the right show little change. The decline over the five trading days before disclosure is consistent with some adjustment before the recorded announcement. Panel B of Appendix Table A.3 reports negative

[6] Appendix Figure A.2 reports the corresponding Fama–French 48-industry paths, which preserve the qualitative patterns.

same-industry departure returns prior to the announcement. Subsequent returns provide no clear evidence of continued losses or a reversal.

One might be concerned that the price movements before disclosure reflect persistent return patterns in the selected locations. For example, technology firms in Austin, Texas, might routinely earn positive abnormal returns even without headquarters news. Appendix Section B.3 explores this possibility by examining placebo returns for incumbent firms and shows that incumbents earn higher abnormal returns around arrival announcements than their average on random dates.

The arrival and departure comparisons remain informative about investors' expectations even when earlier information has already partially been incorporated into prices. If the announcement provides a common update about whether a headquarters will be present, firms with greater expected benefits should gain more upon arrival or lose more upon departure, relative to their existing value. Partial anticipation may attenuate the magnitude of these announcement returns, but it still preserves the relative ordering of gains across more- and less-exposed incumbents. The short window limits exposure to unrelated news while capturing the revision in value around disclosure. Thus, the returns reveal which incumbent groups receive favorable or unfavorable news, even if they do not capture the relocation's full valuation effect. Interpreting these responses as evidence of local spillovers rests on three conditions. First, the gains must reflect benefits from the headquarters' presence, rather than favorable local conditions that would benefit incumbents without the move. Second, unrelated incumbent news must not systematically coincide with the announcement. Third, larger returns must reflect larger expected benefits relative to firm value, rather than differences in how much information was already priced in. The remaining tests therefore examine whether the gains follow the geographic and economic relationships through which spillovers should arise to rule out these alternative explanations.

The remaining analysis focuses on arrivals because the policy question is whether attracting a headquarters creates value for firms already in the destination. The positive arrival response

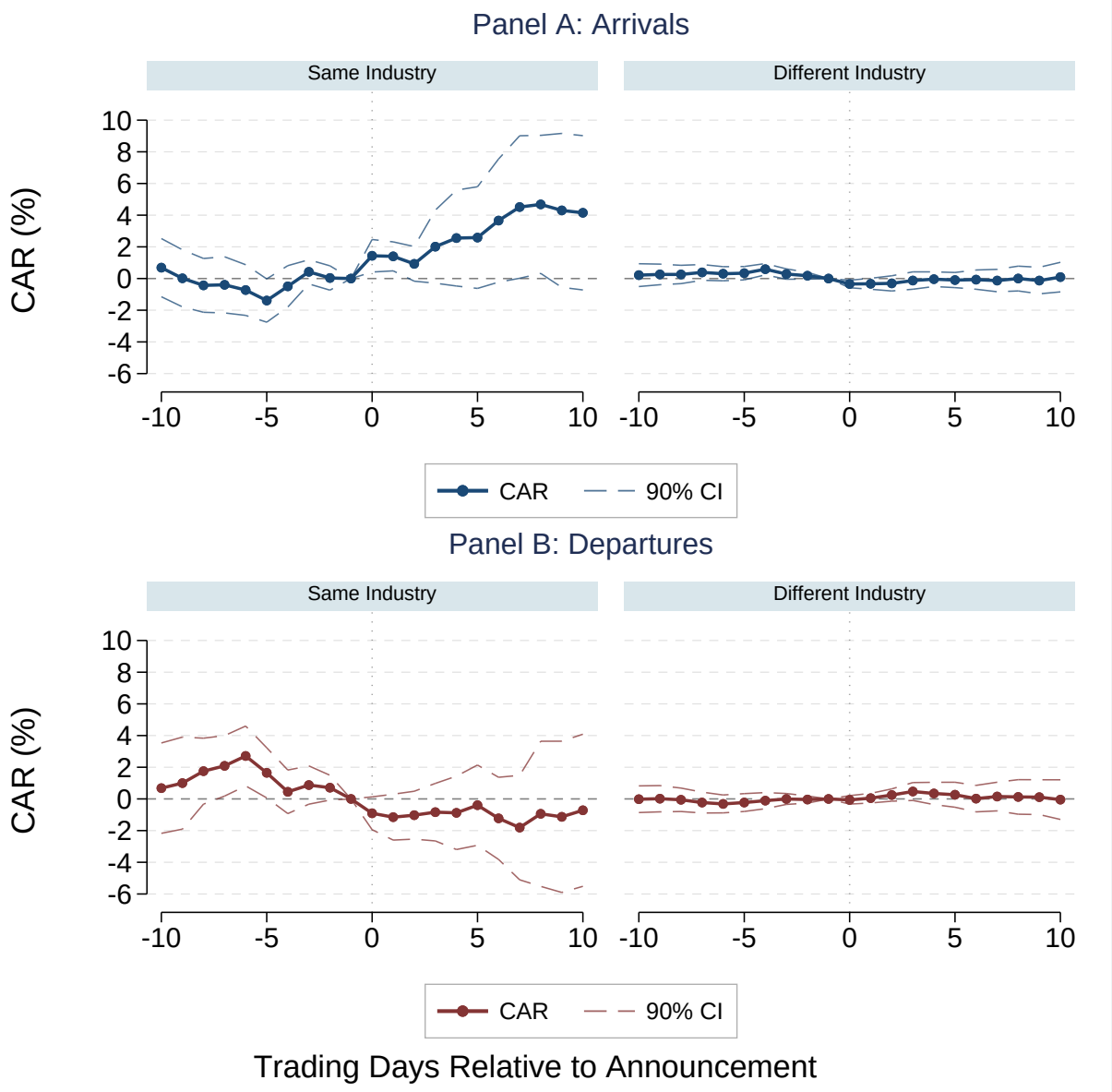


Figure 5. Incumbent Return Paths Around Headquarters Relocation Announcements

Notes: This figure shows mean market-model cumulative abnormal returns, in percent, for public incumbents from ten trading days before through ten trading days after headquarters relocation announcements. Each path is normalized to zero on day −1. Panel A shows destination-side incumbents around arrivals. Panel B shows origin-side incumbents around departures. Same-industry incumbents on the left side of each panel share the relocating headquarters' ZIP code and four-digit NAICS industry. Different-industry incumbents on the right side of each panel share the ZIP code but operate in other four-digit industries. The announcement-window samples contain 81 same-industry and 867 different-industry arrival firm-events, and 31 same-industry and 728 different-industry departure firm-events. Daily paths use available firm-event returns. All panels use the same vertical scale. Solid lines show mean cumulative abnormal returns. Dashed lines trace pointwise 90% confidence bounds using standard errors clustered by relocation announcement and t critical values.

provides evidence on investors' expectations of these benefits. Its concentration among industry peers further suggests that the arriving headquarters' relationship to the existing cluster matters.

The concentration of gains among industry peers raises a further question: does the arrival response depend on how narrowly industry peers are defined? In Figure 6, I compare same-ZIP incumbents that share the arriving firm's four-digit, three-digit, or two-digit NAICS industry. The two-day cumulative abnormal return is 1.40% under the four-digit definition, falls to 0.91% under the three-digit definition, and declines further to 0.56% under the two-digit definition.[7] Returns for different-industry incumbents remain near zero under each definition. Thus, the increase in firm value is largest among incumbents that are most similar to the arriving firm. The steady decline as the industry definition broadens suggests that investors place greater value on arrivals that deepen narrowly defined local clusters.

[7] Under the Fama–French 48-industry model, the corresponding returns are 1.06%, 0.91%, and 0.48%, respectively.

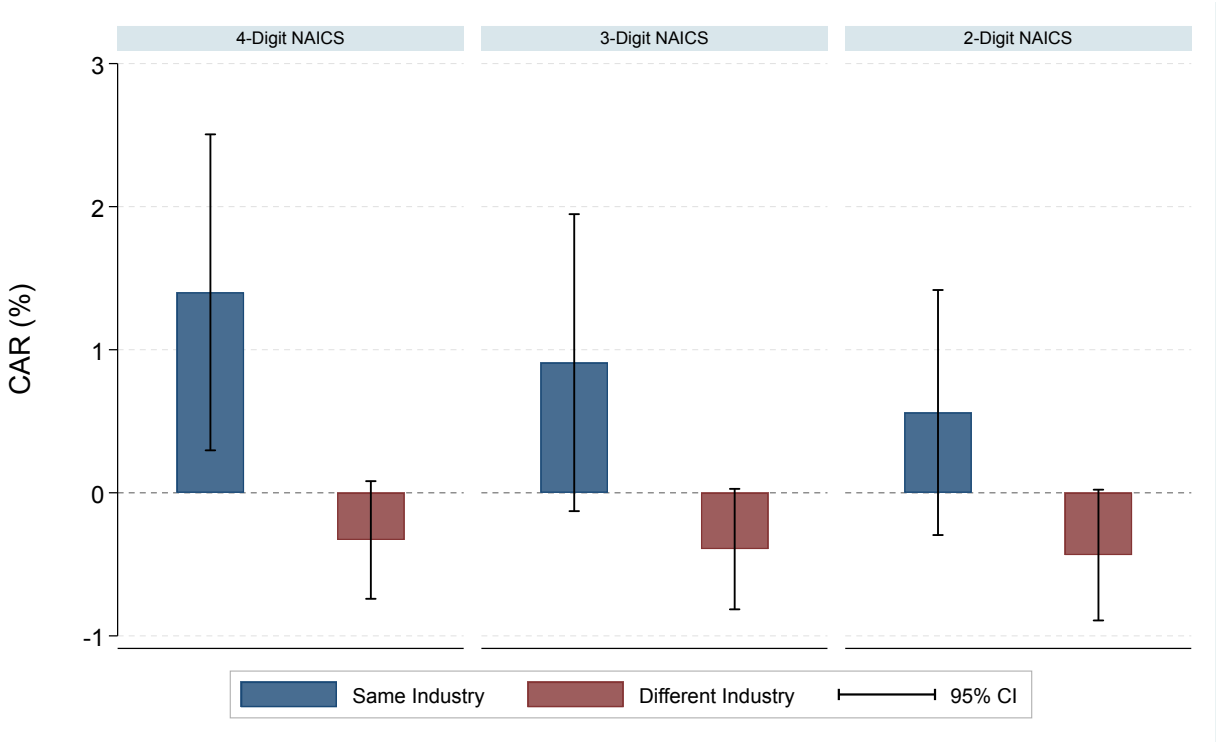


Figure 6. Industry Relatedness and Incumbent Returns

Notes: This figure displays two-day $[0,+1]$ cumulative abnormal returns for destination-side public incumbents located in the arriving headquarters' ZIP code. Incumbents are categorized by whether they share the mover's industry under four-digit ($n = 81$ same-industry and 867 different-industry), three-digit ($n = 127$ and 824), and two-digit ($n = 206$ and 745) NAICS definitions. Whiskers represent 95% confidence intervals using standard errors clustered by relocation announcement.

4 Sources of Same-Industry Returns

The preceding section shows that investors assign value to proximity between firms in the same industry. I next examine the mechanisms that may account for these gains. The urban economics literature emphasizes three channels of agglomeration: labor market pooling across broad commuting areas, input sharing, and localized knowledge spillovers that depend on face-to-face contact (Rosenthal and Strange, 2004, 2020; Jaffe et al., 1993; Matray, 2021; Moretti, 2021). These channels differ in how far their benefits extend and which relationships between firms generate value. I examine these predictions using physical distance, driving time, product similarity, and

supplier links. Because headquarters relocations primarily move executives and research teams rather than physical production, I expect the gains to decline sharply with distance and to be more pronounced among firms with similar products.

4.1 Geographic Reach of the Repricing

This section examines how far the gains from a headquarters arrival extend. This analysis helps distinguish benefits shared across a region from spillovers that depend on close proximity. Knowledge spillovers decline more quickly over short distances than the benefits of regional labor market pooling (Rosenthal and Strange, 2004; Arzaghi and Henderson, 2008; Rosenthal and Strange, 2020; Baum-Snow et al., 2024). The rate at which incumbent returns decline with distance therefore provides a way to distinguish between these mechanisms.

Figure 7 shows how incumbent returns around headquarters arrival announcements vary with linear distance and driving time from the arriving headquarters. Each bar reports the estimated mean market-model cumulative abnormal return for a mutually exclusive proximity group. The estimated group returns are measured relative to zero, rather than as differences from another group. Panel A separates incumbents into three groups: those in the arriving headquarters' ZIP code, those outside that ZIP code but less than three miles away, and those three to ten miles away. Panel B groups incumbents by driving time from the arriving headquarters: at most three minutes, three to nine minutes, and more than nine minutes. Event day zero denotes the announcement trading day. The distance and driving-time comparisons report mean market-model cumulative abnormal returns over the two-day $[0, +1]$ and three-day $[-1, +1]$ windows around headquarters arrival announcements.

Positive abnormal returns around headquarters arrivals are concentrated among the nearest incumbents. In the distance analysis, firms in the arriving headquarters' ZIP code earn a two-day cumulative abnormal return of approximately 1.40%. Returns for incumbents outside the

headquarters' ZIP code but within three miles, and for incumbents three to ten miles away, are near zero. The driving-time analysis also places the largest arrival response among the closest firms. Incumbents located at most three minutes' drive from the arriving headquarters earn a two-day cumulative abnormal return of approximately 1.05%. Estimated returns are negative for incumbents more than three minutes away. This geographic concentration suggests that the results are not driven by favorable industry conditions shared broadly across the destination. The rapid decline in headquarters arrival returns with both physical distance and driving time is also more consistent with highly localized learning than with labor market pooling across a broader region.

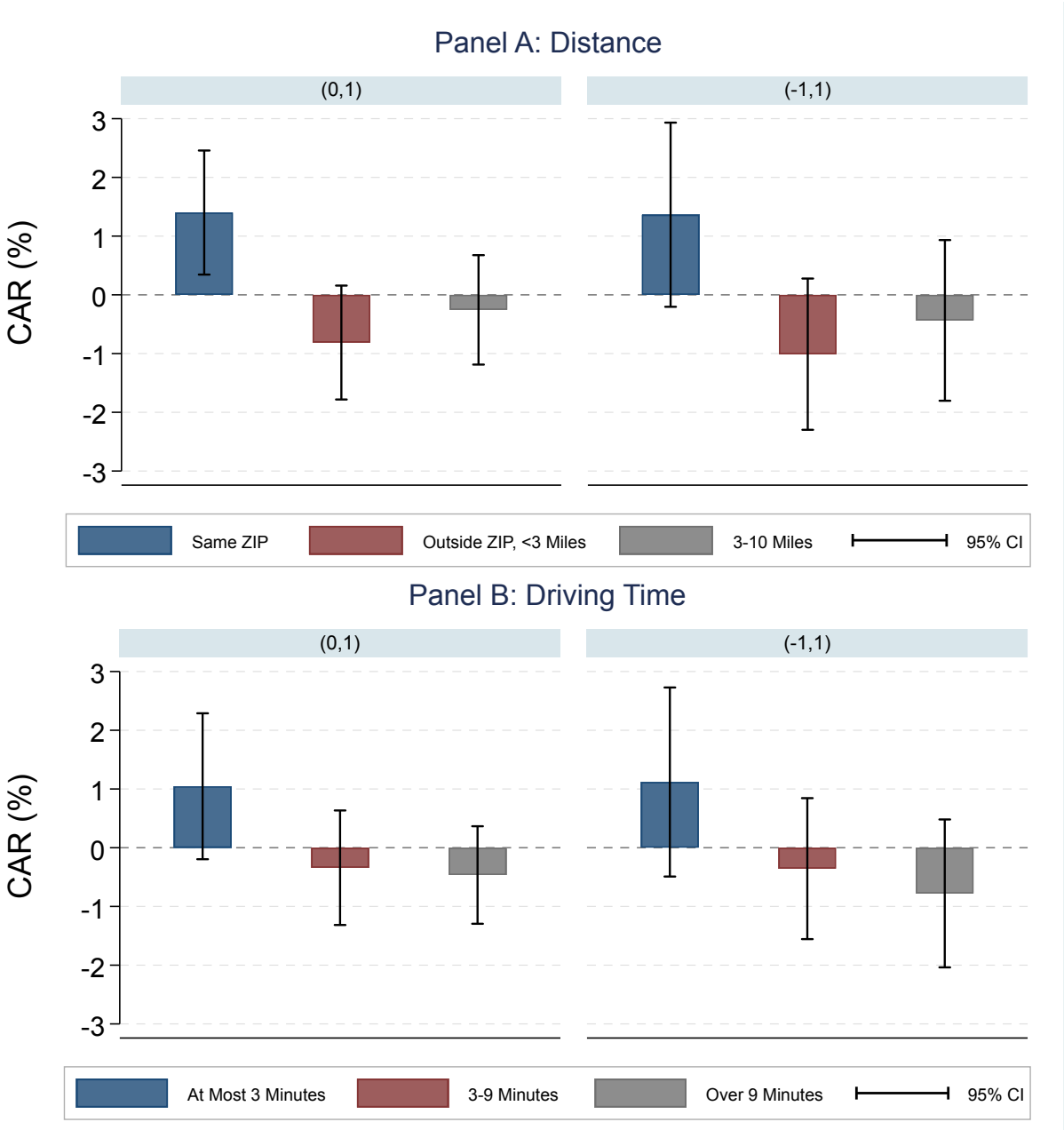


Figure 7. Geographic Proximity and Incumbent Returns

Notes: The geographic analysis plots equal-weighted mean market-model cumulative abnormal returns around headquarters arrival announcements for destination-side public incumbents. The sample includes firms in the arriving headquarters' four-digit North American Industry Classification System (NAICS) industry located less than ten miles away. Panel A groups incumbents by physical distance from the arriving headquarters. The distance groups include same-ZIP ($n = 81$), outside the ZIP but under three miles ($n = 116$), and three to under ten miles ($n = 203$). Panel B groups incumbents by driving time from the arriving headquarters. The driving-time groups include at most three minutes ($n = 94$), three to nine minutes ($n = 90$), and over nine minutes ($n = 216$). Bar heights report estimated mean cumulative abnormal returns for each proximity group. Whiskers represent 95% confidence intervals for the estimated mean cumulative abnormal returns.

4.2 Economic Relatedness and Repricing

I examine whether the positive return response to headquarters arrivals is stronger among nearby incumbents whose products resemble those of the arriving firm. The stock-market gains from headquarters arrivals are already concentrated among geographically close incumbents. Economic relatedness may also matter. Firms operating in similar technological or product areas should be better able to apply knowledge from the arriving headquarters. Consistent with a role for knowledge transfer, Matray (2021) finds that local innovation spillovers are stronger among technologically related firms. If knowledge transfer contributes to headquarters arrival gains, same-ZIP incumbents should earn larger announcement returns when their products more closely resemble those of the arriving firm.

I measure product similarity between same-ZIP incumbents and arriving firms using the Text-Based Network Industry Classification (TNIC) measure developed by Hoberg and Phillips (2016). I form groups of incumbent–mover pairs at or above the 10th, 30th, and 50th percentiles of the positive TNIC similarity distribution. I report a mean cumulative abnormal return for each product-similarity group. Raising the product-similarity threshold progressively excludes less-related incumbent–mover pairs, concentrating the headquarters arrival analysis on firms with more similar products.

Figure 8 shows that estimated headquarters arrival returns increase with product similarity between the arriving headquarters and the incumbent firm. Over the two-day $[0, +1]$ window, mean cumulative abnormal returns are 0.78%, 1.01%, and 1.38% for incumbent–mover pairs at or above the 10th, 30th, and 50th percentiles of positive TNIC similarity, respectively. The three-day $[-1, +1]$ window produces higher estimates of approximately 1.12%, 1.24%, and 1.77% at the corresponding 10th-, 30th-, and 50th-percentile thresholds. The larger estimated gains and statistically positive returns among the most product-similar incumbents are consistent with investors expecting greater benefits when nearby firms can more readily use knowledge from the arriving

headquarters. Therefore, the value of attracting a headquarters depends on how closely the arriving firm's capabilities complement those of firms already operating in the local cluster.

This also suggests the repricing is not merely a region-wide signal. Together, Figure 7 and Figure 8 show that estimated gains are concentrated among nearby incumbents and increase with product similarity to the arriving firm. These patterns weigh against an explanation based only on favorable conditions shared broadly across the destination. Unrelated news or differences in prior information would also need to account for both patterns to explain the return differences. The evidence therefore strengthens the interpretation that investors value local spillovers from the arriving headquarters.

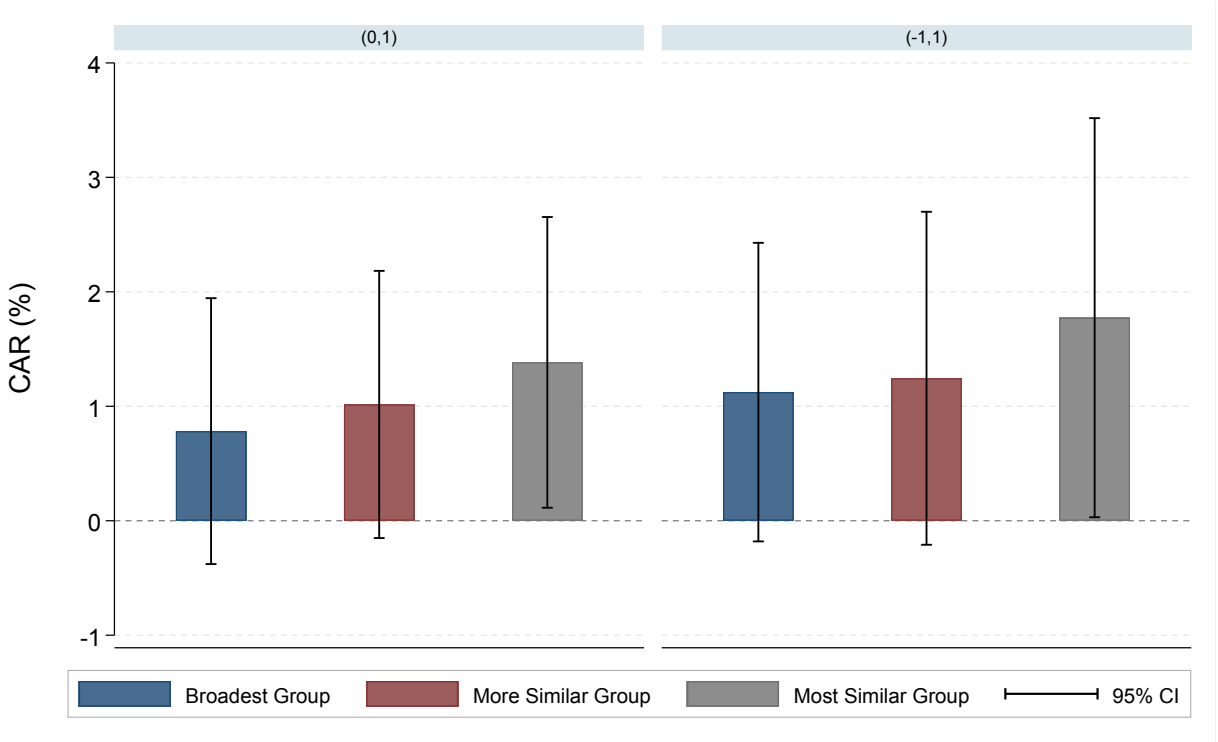


Figure 8. Product Similarity and Incumbent Returns

Notes: The product similarity analysis plots mean market-model cumulative abnormal returns around headquarters arrival announcements for destination-side public incumbents in the arriving headquarters' ZIP code. Incumbents are grouped by Text-Based Network Industry Classification (TNIC) product similarity with the arriving firm. The similarity groups include incumbents at or above the 10th percentile ($n = 110$), 30th percentile ($n = 86$), and 50th percentile ($n = 61$) of product similarity. Bar heights report estimated mean cumulative abnormal returns for each product-similarity group. Whiskers represent 95% confidence intervals for the estimated mean cumulative abnormal returns.

4.3 Repricing Among Supplier-Linked Firms

Local supplier relationships could also explain why headquarters arrival gains are concentrated among nearby firms. Although the distance and product-similarity results are consistent with knowledge exchange, proximity to suppliers and customers can reduce shipping costs (Ellison et al., 2010). Headquarters also benefit from specialized local business services (Davis and Henderson, 2008). The geographic concentration of headquarters arrival returns therefore cannot, by itself, distinguish knowledge spillovers from benefits associated with local supplier relationships.

Headquarters relocations may have limited effects on physical input flows because managerial and strategic functions can be located separately from production (Duranton and Puga, 2005; Davis and Henderson, 2008). The separation of headquarters from production motivates an emphasis on local knowledge exchange since the headquarters brings managerial talent. Consequently, gaining a new headquarters should stimulate local idea exchange more than it alters physical, shipping-intensive input flows. Evidence from U.S. manufacturing also indicates that knowledge spillovers have become more important over time, while input-output linkages have become less important in explaining agglomeration (Steijn et al., 2022).

I test the supplier channel by examining local incumbents in sectors that rely heavily on inputs from the arriving firm's sector. Under this interpretation, the arriving headquarters represents a potential upstream supplier, and the local incumbent is a downstream customer. If proximity to input suppliers drives the valuation gains, these downstream incumbents should earn positive abnormal returns when a headquarters from their primary upstream supplier sector arrives nearby.

I classify an incumbent as supplier-linked when its sector's strongest upstream input relationship is with the arriving headquarters' sector. I measure sector-level production relationships using the Bureau of Economic Analysis (BEA) input-output accounts. For each incumbent's two-digit North American Industry Classification System (NAICS) sector, I identify its strongest upstream-dependency sector and determine whether it matches the arriving headquarters' two-digit sector.

I compare announcement returns for supplier-linked incumbents in the arriving headquarters' ZIP code with returns for incumbents in the same ZIP code and two-digit NAICS industry as the arriving firm.

Figure 9 shows that supplier-linked incumbents do not exhibit the positive valuation response to headquarters arrivals observed among same-industry peers. The supplier-linked mean cumulative abnormal return is -0.72% over the $[0, +1]$ window and -0.35% over the $[-1, +1]$ window. Neither supplier-linked estimate is statistically significant at conventional levels. For comparison, same-ZIP incumbents in the arriving firm's two-digit NAICS industry earn positive cumulative abnormal returns of 0.56% over the $[0, +1]$ window and 0.28% over the $[-1, +1]$ window. Therefore, supplier-linked returns provide little evidence that proximity to upstream input suppliers explains the positive incumbent response to headquarters arrivals.

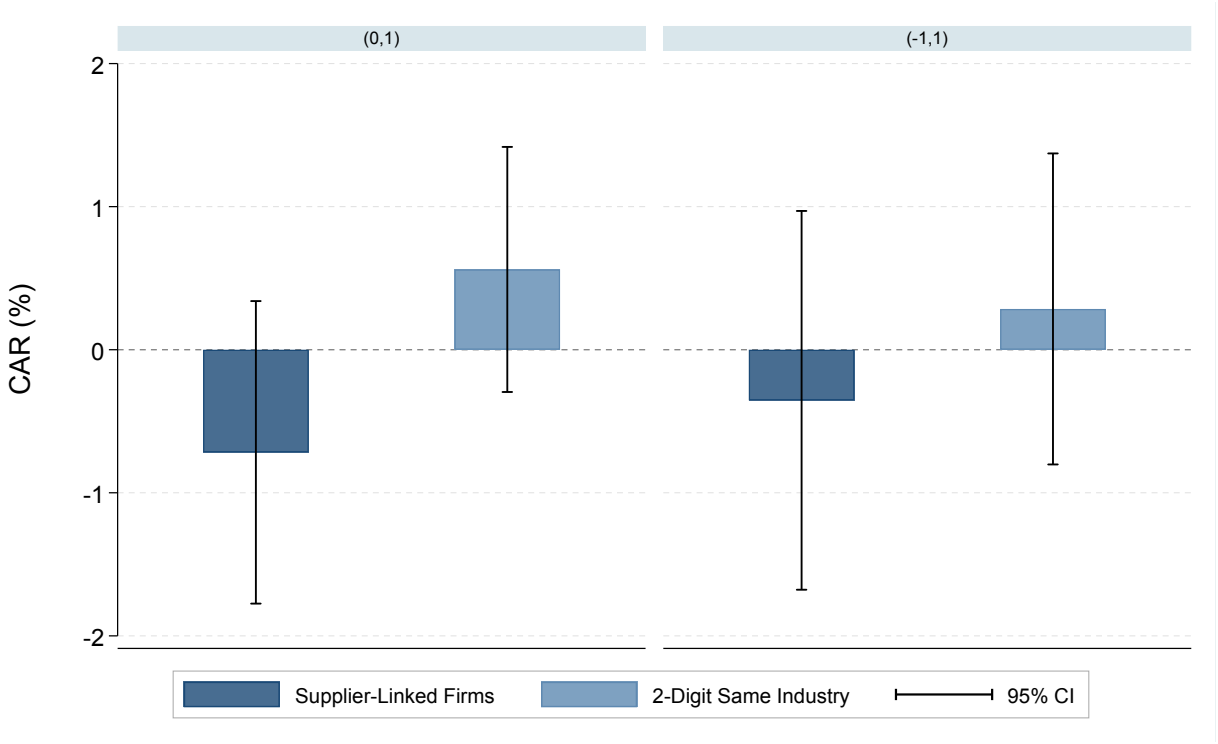


Figure 9. Supplier Links and Incumbent Returns

Notes: The supplier analysis plots cumulative abnormal returns around headquarters arrival announcements for supplier-linked and same-industry destination-side incumbents in the arriving headquarters' ZIP code. Supplier-linked incumbents belong to sectors whose strongest upstream input relationship is with the arriving headquarters' sector. The sample includes supplier-linked incumbents ($n = 110$) and a same-ZIP two-digit industry benchmark for comparison ($n = 206$). Whiskers represent 95% confidence intervals for the estimated mean cumulative abnormal returns.

Across these tests, the gains in firm value are concentrated among geographically close and economically related incumbents, while the supplier-linked estimates are not statistically distinguishable from zero. Therefore, the combined evidence is largely consistent with localized knowledge exchange as a source of value for incumbent firms.

5 Innovation Responses

This section examines whether headquarters arrivals are followed by greater patenting among nearby incumbents. The stock return pattern suggests that investors expect local firms to benefit

from proximity to an arriving industry peer. If knowledge spillovers contribute to these valuation gains, incumbents in the arriving headquarters' industry should produce more patents after the move.

Evaluating how incumbent patenting responds to a headquarters arrival requires addressing the fact that firms do not relocate at random. Moving headquarters may select metropolitan statistical areas (MSAs) where innovation in their industry is already accelerating, or they may target specific neighborhoods with prolific incumbent inventors or improving local infrastructure. To separate local knowledge spillovers from this endogenous sorting, I combine within-MSA peer comparisons with fixed effects and cross-industry placebo tests. Specifically, I compare incumbents in the arrival ZIP code against same-industry peers elsewhere in the same MSA and test whether patenting gains extend to neighboring incumbents in other industries.

5.1 Measuring Local Innovation Spillovers

To test for local innovation spillovers, I examine whether public incumbents in the mover's destination ZIP code and four-digit NAICS industry produce more patents after a headquarters arrives. To separate patent changes within the destination ZIP code from broader regional trends, I compare these same-ZIP firms with never-treated public incumbents in the same four-digit NAICS industry and destination MSA that lie at least three miles (or ten miles) from every headquarters arrival. To the extent that positive knowledge spillovers extend beyond three or ten miles to reach these comparison firms, the estimated difference will understate the true local response.

To handle the staggered timing of headquarters relocations, I use two difference-in-differences methods that avoid the negative weighting problem inherent in standard two-way fixed effects. First, because annual patent counts are nonnegative, highly skewed, and often zero, I estimate the Poisson pseudo-maximum likelihood (PPML) event-study model of Wooldridge (2023).[8] Poisson

[8]Cohn et al. (2022) show that linear regressions of $\log(1 + \text{patents})$ lack a natural interpretation and can yield estimates with the wrong sign.

handles zero counts naturally without transforming the outcome variable, and recent studies of firm-level innovation likewise rely on Poisson specifications for patent counts.[9] Second, alongside the Poisson estimates, I report the linear estimator of Callaway and Sant'Anna (2021) (C&S) in patent-count levels.

Both estimators report treatment effects in numbers of patents, but they make different assumptions about how patenting would have evolved without a headquarters arrival. The Poisson estimator models percentage changes in expected patent counts. Because it measures percentage growth, Poisson treats an increase from 1 to 2 patents (a 100% jump) as a much larger response than an increase from 101 to 102 patents (less than a 1% change). The C&S estimator instead models simple level changes in patent counts. It treats an increase from 1 to 2 patents the same as an increase from 101 to 102 patents, since both equal an addition of one patent.

A two-period example shows how the Poisson and C&S estimators predict different patent counts without an arrival. Consider Atlas Instruments, a hypothetical incumbent in Boston. Suppose an industry peer moves into Atlas's ZIP code in 2015, and Atlas's annual patenting rises from 20 patents in 2014 to 40 patents in 2016. Over the same period, average patenting among distant comparison peers in Boston rises from 50 to 60 patents. This comparison group increase represents either a 20% growth rate or a flat addition of 10 patents. The Poisson model assumes that, without the arrival, Atlas would have matched the comparison group's 20% growth rate. Applying a 20% increase to Atlas's 2014 count of 20 predicts $20 \times 1.2 = 24$ patents in 2016, implying a spillover gain of $40 - 24 = 16$ patents.[10] The C&S estimator instead assumes that Atlas would have matched the comparison group's 10-patent increase. Adding 10 patents to Atlas's 2014 count predicts $20 + 10 = 30$ patents in 2016, implying a spillover gain of $40 - 30 = 10$ patents.

In the full panel of firms and years, the Poisson estimation proceeds in two steps by first estimating proportional effects and then converting those effects into patent counts. First, I model the

[9] See, for example, Sautner et al. (2023) and Hémous et al. (2025).

[10] This single-firm illustration sets Atlas's 2014 count equal to its cohort baseline; in the full Poisson regression with cohort fixed effects, the baseline is estimated at the arrival-cohort level rather than from each individual firm's initial count.

conditional mean of annual patents for firm i in calendar year t as

$$\mathbb{E}[\text{Patents}_{i,t} \mid g(i), t] = \exp\left\{ \alpha_{g(i)} + \lambda_t + \sum_{g>0} \sum_{r=1}^{3} \delta_{g,r} D_{i,t}^{g,r} + \sum_{g>0} \sum_{r=-3}^{-2} \rho_{g,r} D_{i,t}^{g,r} \right\}. \tag{7}$$

Here, the function $g(i)$ assigns firm i to its first-arrival cohort, defined by the calendar year of its first headquarters arrival, with $g(i) = 0$ for never-treated firms. The index $r = t - g$ measures years relative to the arrival for cohort g. Cohort fixed effects $\alpha_{g(i)}$ account for permanent differences across arrival cohorts, and year fixed effects λ_t account for proportional changes in expected patenting common to the sample, such as a nationwide surge in patenting. The indicator $D_{i,t}^{g,r}$ equals one when firm i belongs to cohort g and calendar year $t = g + r$, and zero otherwise. The year immediately before the arrival ($r = -1$) serves as the omitted reference year. I exclude the relocation year ($r = 0$) from the estimation window because headquarters moves occur throughout the calendar year, making it unclear how much of that year's patenting occurred after the arrival. The coefficients $\delta_{g,r}$ and $\rho_{g,r}$ capture cohort-specific differences after and before the move relative to the year prior to the arrival ($r = -1$). These coefficients enter the log conditional mean, while the tables report the implied changes in expected annual patent counts. In the Atlas illustration, $\hat{\delta}_{2015,1} = \ln(40/24) \approx 0.51$, corresponding to an increase of $100[\exp(\hat{\delta}_{2015,1}) - 1] = 66.67\%$.

Equation (7) accounts for permanent differences across firms, cities, and industries. This happens through the cohort and year fixed effects because every treated firm in cohort g shares a treatment indicator ($D_{i,t}^{g,r}$). Therefore, it is not necessary to include separate firm, MSA, or industry fixed effects. In fact, in this Poisson model, including MSA or industry fixed effects would not change the coefficient of interest because the regressors vary only by cohort and year. Thus, the maximum likelihood depends on the number of patents produced by each cohort in each year, not on how those patents are distributed across individual firms within the cohort. To illustrate this, consider again our Atlas example where the 2015 cohort contains Atlas in Boston, with 20

patents in 2014, and an incumbent in Dallas with 5. Atlas may patent more because of its larger R&D staff, Boston's universities, or its industry. A cohort fixed effect fits the cohort's combined 25 patents. Firm, MSA, or industry fixed effects split those patents into 20 for Atlas and 5 for the Dallas firm, but the total stays 25. Because $\delta_{g,r}$ measures growth from this total, every choice yields the same estimate when each cohort's firms appear in the same years. In my unbalanced panel, the estimated average increases in patenting are positive and statistically significant at the 5% level or higher across the cohort, firm, and MSA fixed-effect specifications.

Second, I convert the estimated proportional effects into patents. For each exposed firm-year with an identified post-arrival coefficient, I predict patenting with that coefficient and with it set to zero:

$$\widehat{\Delta \text{Patents}}_{i,t} = \underbrace{\exp(\hat{\alpha}_{g(i)} + \hat{\lambda}_t + \hat{\delta}_{g(i),r})}_{\text{predicted with arrival}} - \underbrace{\exp(\hat{\alpha}_{g(i)} + \hat{\lambda}_t)}_{\text{predicted without arrival}} . \tag{8}$$

Here, $\widehat{\Delta \text{Patents}}_{i,t}$ is the number of patents the arrival adds for exposed firm i in year t. In the Atlas illustration, the predictions for 2016 are 40 and 24 patents, implying a gain of 16 patents. Suppose the predictions without an arrival are 24, 27, and 30 patents in 2016 through 2018, while the predictions with an arrival are 40, 45, and 50 patents. The estimated annual gains are 16, 18, and 20 patents, which average to 18 additional patents per year. Column [1] of Table 3 reports this average of $\widehat{\Delta \text{Patents}}_{i,t}$ across all exposed firm-years in the three years after the move. In the text, I also express this gain as a percentage by dividing the total predicted additional patents by the total predicted patents without an arrival, which equals $100(16+18+20)/(24+27+30) = 66.67\%$ in the illustration.

Column [2] of Table 3 reports the additive group-time difference-in-differences estimator of Callaway and Sant'Anna (2021), which compares level changes in patent counts for each arrival cohort with those of never-treated firms. For each treated cohort g and post-relocation calendar year t, I estimate

$$\tau^{CS}_{g,t} = \mathbb{E}[\text{Patents}_{i,t}(g) - \text{Patents}_{i,t}(0) \mid g(i) = g], \tag{9}$$

where $\text{Patents}_{i,t}(g)$ is firm i's potential patenting outcome in calendar year t when first exposed in cohort g, and $\text{Patents}_{i,t}(0)$ is its potential patenting outcome in the absence of exposure. The parameter $\tau^{CS}_{g,t}$ captures the average treatment effect on firms treated in cohort g during calendar year t. For each cohort g and comparison year t, I first calculate each firm's change in patents from the year before the arrival $(g-1)$ to year t. For Atlas, I subtract its 2014 patent count from its 2016 count $(40-20=20)$. Subtracting each firm's own prior patenting removes fixed additive differences in patent output driven by Atlas's R&D staff size or its location in Boston. In Column [2], I then subtract the average 2014-to-2016 patent change among all eligible never-treated firms in the sample to remove patent-count changes shared across the economy over those years. In the illustration, the comparison firms' increase is $60-50=10$, so the estimated gain is $20-10=10$ patents. Because taking within-firm differences and comparing across the same calendar years removes permanent level differences and common time shocks automatically, this calculation estimates no separate firm, cohort, or year fixed-effect parameters.[11]

Across both specifications, I average the cohort-year treatment effects across the three post-relocation years and cluster standard errors by incumbent ZIP code. Clustering by ZIP code accounts for the possibility that patenting shocks persist within Atlas over time or simultaneously affect several neighboring incumbents sharing Atlas's ZIP code across multiple years, while treating different ZIP codes as independent clusters.

5.2 Innovation Within and Across Industries

Figure 10 plots the yearly event-study coefficients before and after a headquarters arrival. The left panel displays the Poisson estimates with cohort and year fixed effects, and the right panel displays the C&S estimates. Under both distance restrictions, the estimates turn positive after the arrival and remain positive for at least three years. Before the arrival, three of the four series have

[11] In the main patenting tables, I report "Yes" when a specification separately estimates the indicated fixed effects and "No" otherwise. The C&S calculation removes permanent firm characteristics through within-firm changes without separately estimating firm, cohort, or year fixed-effect parameters.

jointly insignificant pre-arrival coefficients for the three-mile Poisson specification and for both C&S specifications. In the ten-mile Poisson specification, treated firms trend upward before the move, and the coefficient three years before the relocation is statistically significant. This trend limits a causal reading of the ten-mile Poisson estimate, but it does not drive the main result. The three-mile sample yields nearly the same post-arrival response.

Table 3 averages these yearly effects over the three years after the arrival and shows that patenting increased for same-industry peer firms that received a new HQ firm in their ZIP code. The controls in Panel A are same-industry incumbents in the same MSA that are at least three miles away from the new HQ. Panel B reports results for control incumbents that are at least ten miles from the relocating HQ. Across specifications, the average annual patenting response for local peers is positive.[12] In Column [1], the Poisson model with arrival-cohort and year fixed effects implies an increase of about 24 annual patents in the three-mile sample and 26 patents in the ten-mile sample, both significant at the 5% level. The fact that the three-mile estimate is nearly as large as the ten-mile estimate suggests that the pre-arrival trend in the ten-mile sample alone does not explain the increase in patenting activity. Relative to the patent counts predicted by the model without a headquarters arrival, these estimates represent increases of 70% and 81%, respectively.

[12] Observation counts differ across columns because Poisson drops FE groups in which every observation has zero patents.

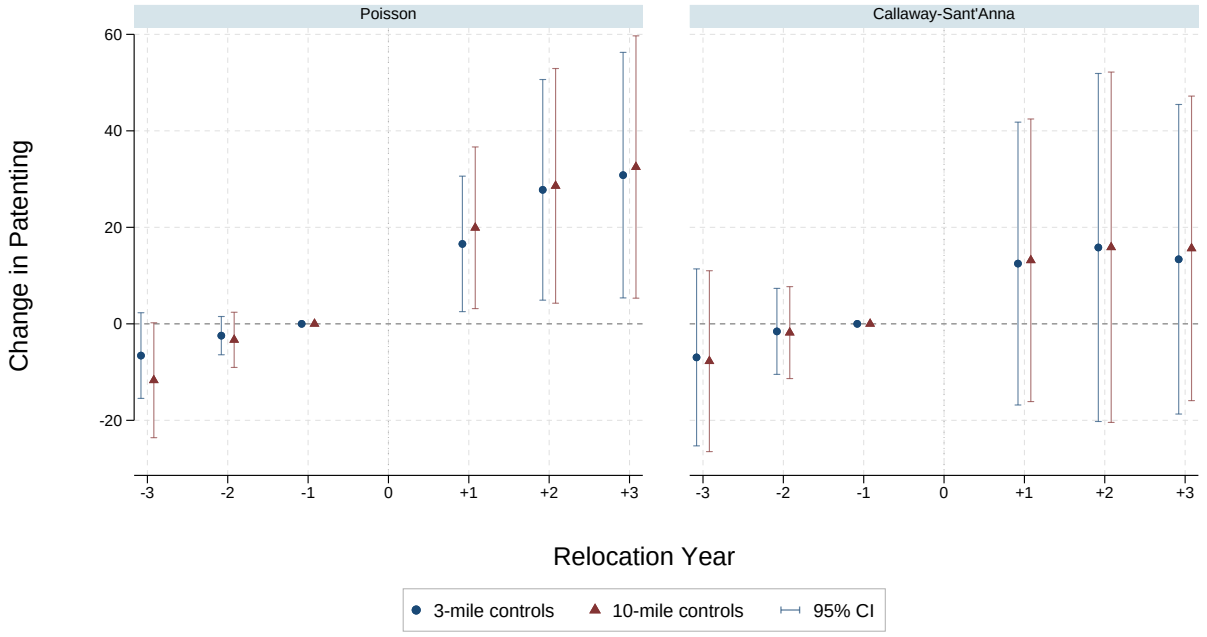


Figure 10. Incumbent Patenting After Headquarters Arrivals

Notes: This figure plots yearly changes in annual patenting for same-ZIP, same-industry public incumbents. The left panel presents Poisson event-study difference-in-differences estimates with cohort and calendar-year fixed effects following Wooldridge (2023). The right panel displays Callaway and Sant'Anna (2021) estimates. Blue circles represent the three-mile comparison group. Red triangles indicate the ten-mile comparison group. The year prior to the relocation is the reference period, and the relocation year is omitted. Whiskers represent 95% confidence intervals using standard errors clustered by incumbent ZIP code.

Column [2] of Table 3 reports the additive C&S estimates. In Column [2], comparing within-firm patent changes against never-treated peers implies increases of about 14 to 15 annual patents.[13] While the coefficients are all positive, the additive estimates are not statistically significant at conventional levels. This lack of precision in the linear model is not surprising given that annual patenting is highly right-skewed with a large mass at zero, and linear OLS estimators in levels are sensitive to extreme counts, which inflates standard errors (Cohn et al., 2022). Nonetheless, both the Poisson and additive estimators point to positive post-arrival patenting gains.

[13]Controlling for MSA-specific patent growth yields nearly identical gains of about 14 and 16 patents, suggesting that city-wide trends do not drive the results.

Table 3. Incumbent Patenting After Headquarters Arrivals

Notes: This table reports average changes in annual patent production over the three years following a headquarters arrival (2004 to 2024), excluding the relocation year. Exposed firms are public incumbents in the arriving headquarters' ZIP code and four-digit NAICS industry. Comparison firms are never-treated same-industry peers in the destination MSA located at least three miles (Panel A) or ten miles (Panel B) away. Column [1] reports Poisson estimates with cohort and calendar-year fixed effects (Wooldridge, 2023). Column [2] reports linear group-time difference-in-differences estimates (Callaway and Sant'Anna, 2021). Standard errors clustered by incumbent ZIP code are in parentheses. Significance at 1%, 5%, and 10% is denoted by ***, **, and *, respectively.

	Poisson	Callaway–Sant'Anna
	[1]	[2]
Panel A: Three-mile comparison group		
Change in annual patents	23.9**	13.9
	(10.0)	(16.5)
Obs.	5,148	5,253
Panel B: Ten-mile comparison group		
Change in annual patents	26.0**	14.8
	(11.2)	(16.4)
Obs.	3,291	3,396
Year FE	Yes	No
Cohort FE	Yes	No

One might be concerned that incumbent firms anticipate a headquarters arrival upon its public announcement and begin expanding their research before the physical move occurs. If a relocation is announced the year before the move, this early response could elevate patenting in the reference year ($r = -1$) and attenuate the estimated post-arrival gain. To address this concern, Internet Appendix Table A.9 restricts the sample to relocations announced and first observed in a destination 10-K in the same calendar year. In addition, Internet Appendix Table A.8 includes the relocation year ($r = 0$) in the post-arrival window rather than omitting it. Consistent with Table 3, both tests show positive post-arrival patenting responses for same-industry incumbents.

I next test whether the positive patenting response extends to nearby incumbents outside the arriving headquarters' industry. This test helps separate industry-specific knowledge spillovers from

neighborhood-wide improvements. Returning to the Boston example, suppose a software headquarters moves into Atlas's ZIP code because the city just upgraded local transit or commercial infrastructure. If those neighborhood improvements encourage innovation across all local businesses, Atlas (a medical instruments manufacturer) should also produce more patents after the software firm arrives, even though the two firms operate in different four-digit NAICS industries. In contrast, if innovation gains depend on sharing knowledge relevant to the arriving firm's industry, a software arrival should not help Atlas design medical instruments, and the positive patenting response should concentrate among same-industry peers.

To test for cross-industry patenting responses, I re-estimate Equations (7) through (9) for same-ZIP incumbents operating in four-digit NAICS industries different from the arriving headquarters. As in the main analysis, I compare these exposed firms against never-treated peers in their own four-digit NAICS industry located at least three or ten miles away in the same MSA. I exclude incumbents that experience an overlapping own-industry arrival.

Figure 11 plots yearly changes in patenting for these different-industry incumbents. The left panel displays the Poisson estimates with cohort and year fixed effects, and the right panel displays the C&S estimates. Prior to the move, the yearly estimates remain close to zero across both models and distance restrictions, and the pre-arrival coefficients are jointly insignificant in all four series. After the headquarters arrives, the yearly estimates show no upward trajectory. Across the three post-arrival years, the point estimates range from -4 to 0 annual patents under the Poisson model and from -3 to 0 patents under the C&S estimator. This pattern indicates that the positive patenting response is concentrated among same-industry peers rather than shared across all firms in the destination ZIP code.

Table 4 averages these yearly effects over the three years after the arrival and shows that incumbents in other industries do not produce more patents when a headquarters enters their ZIP code. Across specifications, the estimated average annual patenting responses for local firms are

negative.[14] In Column [1], the Poisson model with arrival-cohort and year fixed effects implies a decrease of about 3 annual patents in the three-mile sample and 2 patents in the ten-mile sample, neither statistically significant at conventional levels.[15] Column [2] of Table 4 reports the additive C&S estimates. In Column [2], comparing within-firm patent changes against never-treated peers implies decreases of about 1 to 2 annual patents.[16] While the coefficients are all negative, the additive estimates are not statistically significant at conventional levels. Both the Poisson and additive estimators provide no evidence of positive post-arrival patenting gains for incumbents outside the arriving headquarters' industry.

Taken together, the positive same-industry patenting responses in Table 3 and the absence of corresponding gains outside the arriving firm's industry in Table 4 are consistent with industry-specific knowledge spillovers rather than broad neighborhood improvements. Comparing exposed incumbents against peers in the same industry and metropolitan area ensures that national industry surges and city-wide innovation booms do not drive the estimated gains. At the same time, the lack of a patenting response among neighboring firms in other industries rules out hyper-local infrastructure or commercial upgrades that would likely benefit all businesses in the destination ZIP code. Incumbents operating in the arriving firm's industry are uniquely positioned to apply a new neighbor's ideas to their own research, providing a realized innovative counterpart to the stock market gains observed at the announcement. Thus, for policy makers, relocation incentives are most likely to spur local innovation when they attract companies to areas with existing industry peers.

[14] Observation counts differ across columns because Poisson drops FE groups in which every observation has zero patents.

[15] In the alternative specification with firm fixed effects, the Poisson estimates indicate declines of about 13 annual patents in both samples, both significant at the 1% level.

[16] As a check, I regress comparison firms' within-firm patent changes on MSA indicators separately for each cohort-year comparison and subtract the fitted local change from each exposed firm's change. The estimates are still negative.

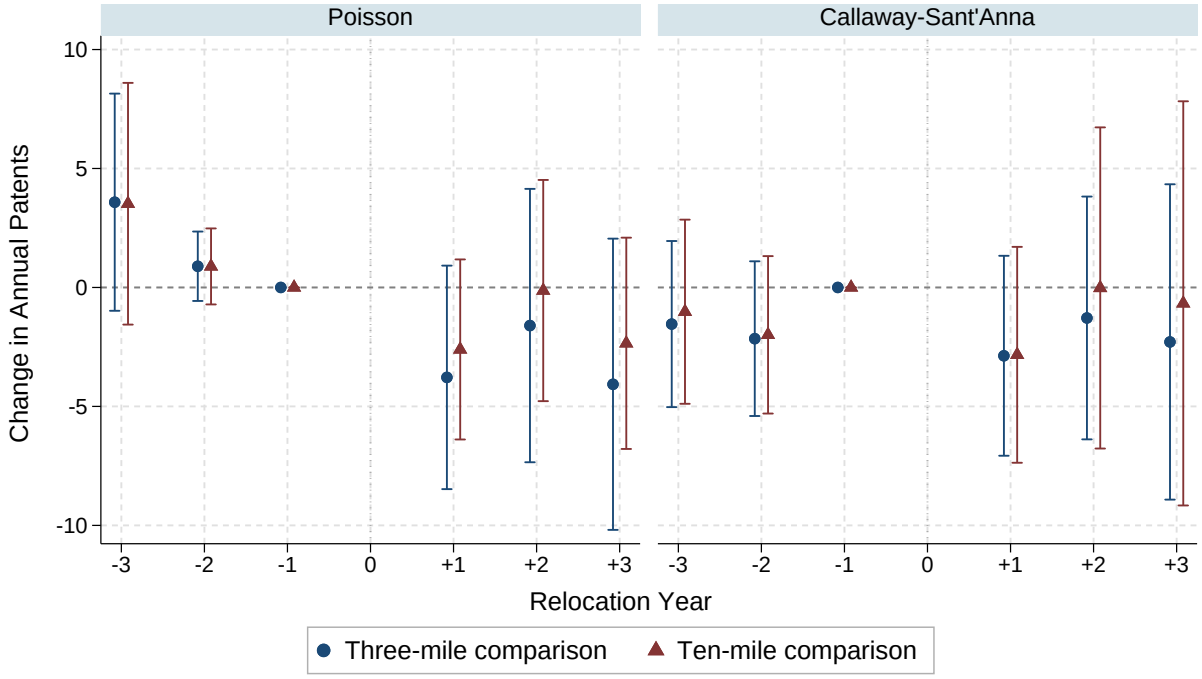


Figure 11. Different-Industry Incumbent Patenting After Headquarters Arrivals

Notes: This figure plots yearly changes in annual patenting for same-ZIP public incumbents in four-digit NAICS industries different from the arriving headquarters. The left panel uses the Poisson estimator of Wooldridge (2023) with cohort and year fixed effects. The right panel uses the additive estimator of Callaway and Sant'Anna (2021). Blue circles represent the three-mile comparison group and red triangles the ten-mile comparison group. The reference period is the year before relocation. The relocation year is omitted. Whiskers represent 95% confidence intervals using standard errors clustered by incumbent ZIP code.

Table 4. Different-Industry Incumbent Patenting After Headquarters Arrivals

Notes: This table reports average changes in annual patent production over the three years following a headquarters arrival (2004 to 2024) for same-ZIP public incumbents in four-digit NAICS industries different from the arriving headquarters. Comparison firms are never-treated peers in the destination MSA located at least three miles (Panel A) or ten miles (Panel B) away. Column [1] reports Poisson estimates with cohort and calendar-year fixed effects (Wooldridge, 2023). Column [2] reports linear group-time difference-in-differences estimates (Callaway and Sant'Anna, 2021). Standard errors clustered by incumbent ZIP code are in parentheses. Significance at 1%, 5%, and 10% is denoted by ***, **, and *, respectively.

	Poisson	Callaway–Sant'Anna
	[1]	[2]
Panel A: Three-mile comparison group		
Change in annual patents	-3.1	-2.2
	(2.7)	(2.4)
Obs.	22,634	22,637
Panel B: Ten-mile comparison group		
Change in annual patents	-1.7	-1.3
	(2.1)	(3.0)
Obs.	12,589	12,622
Year FE	Yes	No
Cohort FE	Yes	No

6 Alternative Interpretations

The preceding sections show that same-industry incumbents gain market value when a headquarters arrival is announced and produce more patents after the move occurs. While these patterns are consistent with localized knowledge spillovers, several alternative explanations could account for the positive announcement returns. However, any competing explanation must simultaneously account for the concentration of returns within the arriving firm's industry, their rapid decay with distance, the losses at origin locations, and the subsequent increase in realized patenting. In this section, I evaluate whether five alternative mechanisms can explain my results.

Broad local conditions and regional signaling. One concern is that a headquarters relocation simply signals management's private view of favorable regional macroeconomic conditions. If the event merely broadcasted favorable local fundamentals, such as broad tax incentives or new regional infrastructure, the valuation response should logically extend across all industries throughout the metropolitan area. Instead, the market repricing is strictly confined to the relocating firm's specific industry and decays rapidly over short physical distances. This combination of sector isolation and spatial decay suggests that investors are not merely pricing a general macroeconomic signal.

Local investor trading. Another alternative is that the observed returns simply reflect coordinated buying pressure from local retail investors rather than a change in perceived clustering value (Pirinsky and Wang, 2006). However, a broad local-investor channel predicts similar buying pressure across all firms near the destination. I find instead that returns for different-industry firms in the same ZIP code are near zero. The positive response concentrates strictly among same-industry peers and increases with product similarity. Moreover, the subsequent real adjustments in patenting suggest that prices reflect fundamental economic changes rather than transient retail sentiment. The joint evidence points to economic value generated by proximity among related firms rather than local attention generating a short-window repricing.

Bundled information. Another threat to identification is that management strategically times relocation announcements to coincide with positive industry trends or firm-specific disclosures. Yet, for bundled information to explain these results, it would need to convey value primarily to firms very close to the relocating headquarters and attenuate sharply over short distances. Strategic coordination should logically affect same-industry firms across the entire metropolitan area. Nonetheless, to formally address this, Appendix Table A.6 in Appendix Section B.4 drops incumbent firms that filed a Form 8-K within two days of the relocation announcement and shows that arrivals are associated with a positive return while departures are associated with a negative return.

Mergers and acquisitions. The arrival premium might simply represent an increased probability that nearby incumbents will become acquisition targets, or it could capture contemporaneous transaction news (Song and Walkling, 2000). I limit contemporaneous news mechanically by excluding merger-related articles during sample construction. Furthermore, excluding incumbent firm-events with an 8-K filing near the announcement (Internet Appendix Table A.6) preserves the positive point estimates for same-industry arrivals and the negative estimates for same-industry departures. While proximity and relatedness facilitate transactions, a pure takeover-probability mechanism fails to explain the negative returns among peers losing a headquarters at the origin. The subsequent real adjustments in patenting further show that the premium reflects fundamental operational changes, not just a revised probability of future acquisitions.

Economic development incentives and political favors. A final concern is that local governments attract new headquarters using industry-specific political favors, such as tax credits or regulatory waivers, that investors expect will be extended to existing local peers (Slattery and Zidar, 2020; Bartik, 2020). The spatial decay of the valuation premium challenges this political coordination explanation. Because the response region is generally much smaller than the geographic boundaries of a typical municipality or enterprise zone, the localized premium is unlikely to reflect a town-wide or county-wide policy change. Furthermore, a purely financial windfall from a tax abatement cannot easily explain the subsequent changes in real business activity where nearby firms in the same industry increase their patenting. Thus my results point toward a local benefit from being close to firms doing similar work.

7 Conclusion

State and local governments justify corporate attraction subsidies by promising local economic spillovers. Using a sample of U.S. headquarters relocations, I show that equity markets capitalize these expected spillovers into the stock prices of neighboring incumbent firms. The average ex-

posed same-industry incumbent experiences a 1.40% increase in shareholder value over the two-day announcement window. This valuation premium decays rapidly with physical distance and strengthens among firms with similar product descriptions. I find no comparable price reaction for cross-industry firms or for those linked solely through supply chains. These patterns suggest that the expected benefits arise from hyper-local knowledge spillovers between technologically related peers rather than from broad regional labor pooling or input sharing.

More broadly, I show that corporate geography is capitalized beyond the relocating firm itself. A headquarters location decision affects the equity values of economically related firms already operating nearby. For place-based policy, these estimates identify a benefit where gains accrue specifically to narrowly defined local industry clusters. Therefore, relocation incentives are most defensible when they attract firms that deepen existing industry clusters.

References

ARROW, K. J. (1962): "Economic Welfare and the Allocation of Resources for Invention," in The Rate and Direction of Inventive Activity: Economic and Social Factors, ed. by R. R. Nelson, Princeton, NJ: Princeton University Press, 609–626.

ARZAGHI, M. AND J. V. HENDERSON (2008): "Networking off Madison Avenue," Review of Economic Studies, 75, 1011–1038.

BARTIK, T. J. (2020): "Using Place-Based Jobs Policies to Help Distressed Communities," Journal of Economic Perspectives, 34, 99–127.

BARTIK, T. J. AND N. SOTHERLAND (2019): "Realistic Local Job Multipliers," Employment Research, 26, 1–3.

BAUM-SNOW, N., N. GENDRON-CARRIER, AND R. PAVAN (2024): "Local Productivity Spillovers," American Economic Review, 114, 1030–1069.

BERNSTEIN, S., E. COLONNELLI, X. GIROUD, AND B. IVERSON (2019): "Bankruptcy Spillovers," Journal of Financial Economics, 133, 608–633.

BROWN, S. J. AND J. B. WARNER (1985): "Using Daily Stock Returns: The Case of Event Studies," Journal of Financial Economics, 14, 3–31.

CALLAWAY, B. AND P. H. C. SANT'ANNA (2021): "Difference-in-Differences with Multiple Time Periods," Journal of Econometrics, 225, 200–230.

COHEN, L. AND A. FRAZZINI (2008): "Economic Links and Predictable Returns," Journal of Finance, 63, 1977–2011.

COHN, J. B., Z. LIU, AND M. I. WARDLAW (2022): "Count (and Count-Like) Data in Finance," Journal of Financial Economics, 146, 529–551.

COMBES, P. P. AND L. GOBILLON (2015): "The Empirics of Agglomeration Economies," in Handbook of Regional and Urban Economics, Elsevier, vol. 5, 247–348.

DAVIS, J. C. AND J. V. HENDERSON (2008): "The Agglomeration of Headquarters," Regional Science and Urban Economics, 38, 445–460.

DELLAVIGNA, S. AND J. M. POLLET (2009): "Investor Inattention and Friday Earnings Announcements," Journal of Finance, 64, 709–749.

DOUGAL, C., C. A. PARSONS, AND S. TITMAN (2022): "The Geography of Value Creation," Review of Financial Studies, 35, 4201–4248.

DUCHIN, R. AND D. SOSYURA (2021): "Remotely Productive: The Efficacy of Remote Work for Executives," Working Paper.

DURANTON, G. AND D. PUGA (2001): "Nursery Cities: Urban Diversity, Process Innovation, and the Life Cycle of Products," American Economic Review, 91, 1454–1477.

——— (2005): "From Sectoral to Functional Urban Specialisation," Journal of Urban Economics, 57, 343–370.

ELLISON, G., E. L. GLAESER, AND W. R. KERR (2010): "What Causes Industry Agglomeration? Evidence from Coagglomeration Patterns," American Economic Review, 100, 1195–1213.

FAMA, E. F. AND K. R. FRENCH (1997): "Industry Costs of Equity," Journal of Financial Economics, 43, 153–193.

GHOSH, C., M. RODRIGUEZ, AND C. F. SIRMANS (1995): "Gains from Corporate Headquarters Relocations: Evidence from the Stock Market," Journal of Urban Economics, 38, 291–311.

GIROUD, X. (2013): "Proximity and Investment: Evidence from Plant-Level Data," Quarterly Journal of Economics, 128, 861–915.

GLAESER, E. L., H. D. KALLAL, J. A. SCHEINKMAN, AND A. SHLEIFER (1992): "Growth in Cities," Journal of Political Economy, 100, 1126–1152.

GREENSTONE, M., R. HORNBECK, AND E. MORETTI (2010): "Identifying Agglomeration Spillovers: Evidence from Winners and Losers of Large Plant Openings," Journal of Political Economy, 118, 536–598.

GUPTA, R. R. (2020): "Does Goliath Help David? Anchor Firms and Startup Clusters," Working Paper.

HÉMOUS, D., M. OLSEN, C. ZANELLA, AND A. DECHEZLEPRÊTRE (2025): "Induced Automation Innovation: Evidence from Firm-Level Patent Data," Journal of Political Economy, 133, 1975–2028.

HENDERSON, J. V. (2003): "Marshall's Scale Economies," Journal of Urban Economics, 53, 1–28.

HIRSHLEIFER, D., S. S. LIM, AND S. H. TEOH (2009): "Driven to Distraction: Extraneous Events and Underreaction to Earnings News," Journal of Finance, 64, 2289–2325.

HOBERG, G. AND G. PHILLIPS (2016): "Text-Based Network Industries and Endogenous Product Differentiation," Journal of Political Economy, 124, 1423–1465.

HU, M. R., D. TSANG, AND W. X. WAN (2024): "Corporate Relocation and Housing Market Spillovers," Management Science, 71, 4344–4376.

JACOBS, J. (1969): The Economy of Cities, New York: Random House.

JAFFE, A. B., M. TRAJTENBERG, AND R. HENDERSON (1993): "Geographic Localization of Knowledge Spillovers as Evidenced by Patent Citations," Quarterly Journal of Economics, 108, 577–598.

KOGAN, L., D. PAPANIKOLAOU, A. SERU, AND N. STOFFMAN (2017): "Technological Innovation, Resource Allocation, and Growth," Quarterly Journal of Economics, 132, 665–712.

KOTHARI, S. P. AND J. B. WARNER (2007): "Econometrics of Event Studies," in Handbook of Empirical Corporate Finance, Elsevier, 3–36.

MACKINLAY, A. C. (1997): "Event Studies in Economics and Finance," Journal of Economic Literature, 35, 13–39.

MALATESTA, P. H. AND R. THOMPSON (1985): "Partially Anticipated Events: A Model of Stock Price Reactions with an Application to Corporate Acquisitions," Journal of Financial Economics, 14, 237–250.

MARSHALL, A. (1890): Principles of Economics, Macmillan.

MATRAY, A. (2021): "The Local Innovation Spillovers of Listed Firms," Journal of Financial Economics, 141, 395–412.

MORETTI, E. (2021): "The Effect of High-Tech Clusters on the Productivity of Top Inventors," American Economic Review, 111, 3328–3375.

PIRINSKY, C. AND Q. WANG (2006): "Does Corporate Headquarters Location Matter for Stock Returns?" Journal of Finance, 61, 1991–2015.

ROMER, P. M. (1986): "Increasing Returns and Long-Run Growth," Journal of Political Economy, 94, 1002–1037.

ROSENTHAL, S. S. AND W. C. STRANGE (2004): "Evidence on the Nature and Sources of Agglomeration Economies," in Handbook of Regional and Urban Economics, Elsevier, vol. 4, 2119–2171.

——— (2020): "How Close Is Close? The Spatial Reach of Agglomeration Economies," Journal of Economic Perspectives, 34, 27–49.

SAUTNER, Z., L. VAN LENT, G. VILKOV, AND R. ZHANG (2023): "Firm-Level Climate Change Exposure," Journal of Finance, 78, 1449–1498.

SLATTERY, C. AND O. ZIDAR (2020): "Evaluating State and Local Business Incentives," Journal of Economic Perspectives, 34, 90–118.

SONG, M. H. AND R. A. WALKLING (2000): "Abnormal Returns to Rivals of Acquisition Targets: A Test of the Acquisition Probability Hypothesis," Journal of Financial Economics, 55, 143–171.

STEIJN, M. P., H. R. KOSTER, AND F. G. VAN OORT (2022): "The Dynamics of Industry Agglomeration: Evidence from 44 Years of Coagglomeration Patterns," Journal of Urban Economics, 130, 103456.

STRAUSS-KAHN, V. AND X. VIVES (2009): "Why and Where Do Headquarters Move?" Regional Science and Urban Economics, 39, 168–186.

WOOLDRIDGE, J. M. (2023): "Simple Approaches to Nonlinear Difference-in-Differences with Panel Data," Econometrics Journal, 26, C31–C66.

Internet Appendix

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A Variable Definitions

I define product similarity, supplier links, and annual patent counts in Appendix Table A.1. The first two measures distinguish relationships between an arriving headquarters and nearby incumbents. Patent counts measure realized innovation. I assign zero patents to valid Compustat firm-years without a linked patent record.

Appendix Table A.1. Variable Definitions

Notes: This table shows definitions for the main outcomes and exposure measures. Annual patent counts are measured at the firm-year level.

Variable	Description
TNIC Product Similarity	Pairwise similarity between the product descriptions of the relocating headquarters and incumbent firm from the Text-Based Network Industry Classification measure of Hoberg and Phillips (2016). Higher percentile cutoffs retain more product-similar pairs.
Supplier Link	Indicator constructed from the BEA Input–Output Accounts that equals one when the relocating headquarters' two-digit NAICS sector is the primary upstream-dependency sector of the incumbent's two-digit NAICS sector.
Annual Patent Count	Number of patents assigned to a public firm by issue year in the KPSS patent data linked to Compustat (Kogan et al., 2017). A valid Compustat firm-year without a patent record is assigned a count of zero.

B Return Tests

I examine the incumbent-return evidence for headquarters arrivals and departures in Section 3. The tests vary industry definitions, distinguish returns before and after the announcement, and compare announcement returns with returns for the same firms on other dates. I also examine contemporaneous incumbent filings and leave-one-out estimates.

B.1 Return Samples

Appendix Table A.2 reports the firm-event and announcement counts for the return tests.

Appendix Table A.2. Sample Across Return Tests

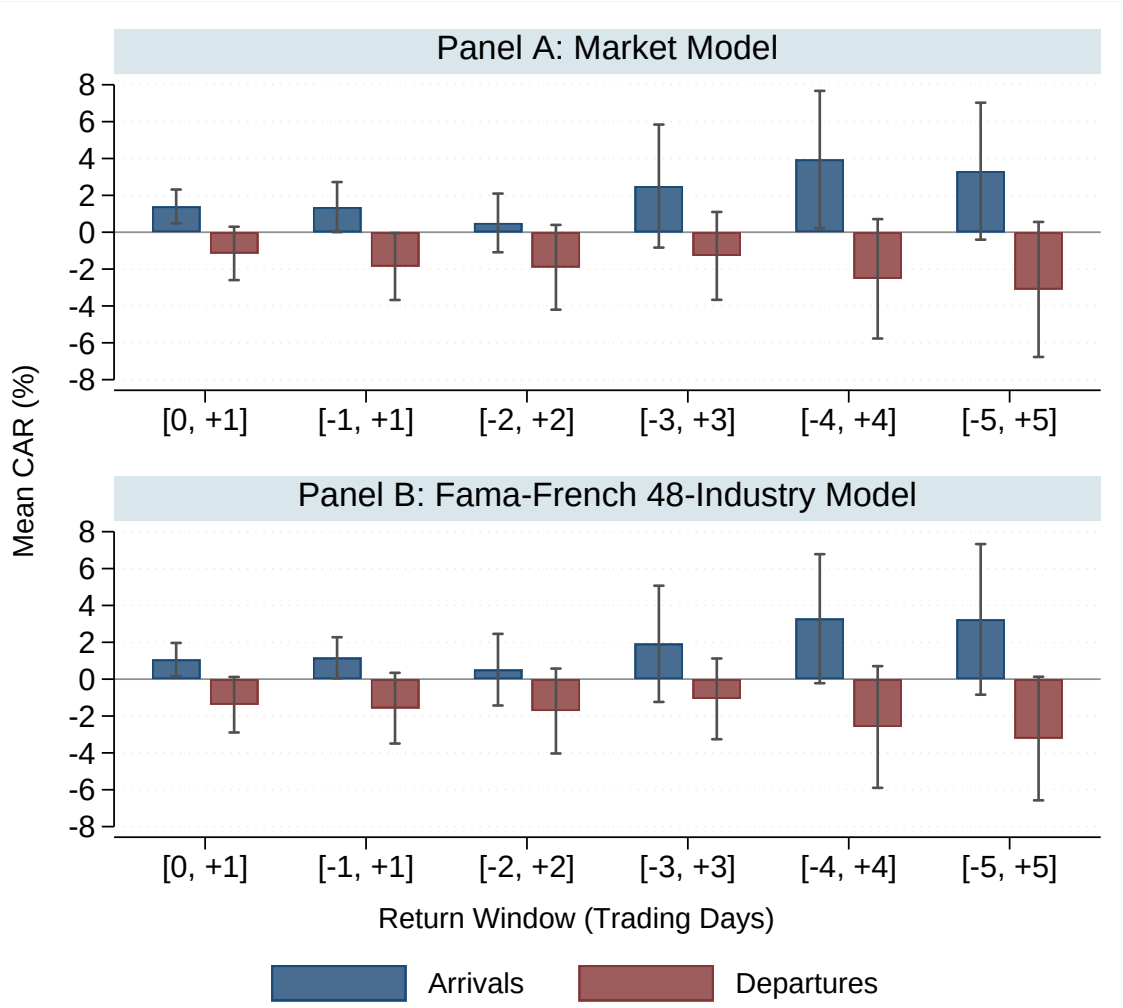
Notes: This table reports the number of incumbent firm-event observations and distinct relocation announcements for each return sample in the main text and appendix, measured over the two-day $[0, +1]$ announcement window under the market model. Dashes indicate tests conducted only for headquarters arrivals.

	Arrivals		Departures	
	Firm-Events	Announcements	Firm-Events	Announcements
Panel A: Industry Definitions				
Four-Digit NAICS, Same Industry	81	24	31	21
Four-Digit NAICS, Different Industry	867	138	728	119
Three-Digit NAICS, Same Industry	127	37	41	26
Three-Digit NAICS, Different Industry	824	136	733	121
Two-Digit NAICS, Same Industry	206	64	107	55
Two-Digit NAICS, Different Industry	745	132	667	116
Panel B: Geographic Proximity				
Same ZIP	81	24	–	–
Outside ZIP, Under 3 Miles	116	26	–	–
3 to Under 10 Miles	203	49	–	–
Driving Time, At Most 3 Minutes	94	26	–	–
Driving Time, Over 3 Through 9 Minutes	90	23	–	–
Driving Time, Over 9 Minutes	216	50	–	–
Panel C: Product and Supplier Links				
TNIC, At or Above 10th Percentile	110	27	–	–
TNIC, At or Above 30th Percentile	86	24	–	–
TNIC, At or Above 50th Percentile	61	17	–	–
Supplier-Linked Incumbents	110	41	–	–
Panel D: Additional Return Tests				
Excluding Contemporaneous 8-K Filings	62	20	25	18
Future Comparison, 12 Months	81	24	30	21
Future Comparison, 18 Months	78	24	29	21
Future Comparison, 24 Months	78	24	25	17

B.2 Announcement Timing and Return Windows

Figure A.1 compares the two-day announcement returns with symmetric windows extending up to five trading days before and after the announcement. Arrival estimates remain positive and depar-

ture estimates remain negative under both models. Thus, the direction of the valuation response is robust to allowing for adjustment around the recorded announcement date.



Appendix Figure A.1. Incumbent Returns Across Announcement Windows

Notes: This figure shows mean cumulative abnormal returns for same-ZIP incumbents in the relocating firm's four-digit NAICS industry. Blue bars report arrivals and red bars report departures. Each window includes all trading days between its displayed endpoints, with day zero denoting the announcement. Panel A uses the market model and Panel B uses the Fama–French 48-industry model. Each window and model includes 81 arrival and 31 departure firm-events. Whiskers represent 90% confidence intervals using standard errors clustered by relocation announcement.

Following Ghosh et al. (1995), who study returns for relocating firms, I examine the return patterns of non-moving incumbents around headquarters arrival and departure announcements. Table A.3 separates returns before, during, and after the announcement for incumbents in the destination ZIP code in Panel A and the origin ZIP code in Panel B. Within each panel, the comparisons distinguish firms in the relocating headquarters' four-digit NAICS industry from firms in other industries. Columns [1]–[3] use the market model, and columns [4]–[6] use the Fama–French 48-industry model. The return calculations and standard errors follow Section 2.2.

For both arrivals and departures, I separate returns before the announcement, over $[-20,-1]$, $[-10,-1]$, and $[-5,-1]$, from returns during the announcement, over $[0,+1]$ and $[-1,+1]$, and afterward, over $[2,10]$, $[11,20]$, and $[2,20]$. Day 0 is the first trading day on or after the recorded announcement. All windows count trading days and include both endpoints.

Appendix Table A.3. Incumbent Returns by Industry Around Headquarters Announcements

Notes: This table reports mean cumulative abnormal returns (in percent) for public incumbents in destination ZIP codes (Panel A) and origin ZIP codes (Panel B) across alternative trading-day windows relative to the announcement date ($t = 0$). Same-industry incumbents share the relocating firm's four-digit NAICS industry, and differences report the same-industry mean minus the different-industry mean in percentage points. Observation counts correspond to the $[0, +1]$ window. Standard errors clustered by relocation announcement are in parentheses. Significance at 1%, 5%, and 10% is denoted by ***, **, and *, respectively.

	Market Model			Industry Model		
	Same	Different	Difference	Same	Different	Difference
	[1]	[2]	[3]	[4]	[5]	[6]
						Panel A: Arrivals
Before $[-20, -1]$	4.167	−0.715	4.883	4.179	−1.151*	5.330
	(2.995)	(0.646)	(2.971)	(3.396)	(0.651)	(3.365)
Before $[-10, -1]$	−0.074	−0.235	0.161	0.808	−0.716*	1.523
	(0.962)	(0.458)	(1.047)	(1.032)	(0.428)	(1.109)
Before $[-5, -1]$	0.726	−0.304	1.030	1.162	−0.559**	1.721*
	(0.933)	(0.269)	(0.905)	(0.973)	(0.245)	(0.942)
Announcement $[0, +1]$	1.401**	−0.330	1.731***	1.063*	−0.190	1.253**
	(0.534)	(0.208)	(0.552)	(0.526)	(0.176)	(0.528)
Announcement $[-1, +1]$	1.364*	−0.509**	1.873**	1.157*	−0.352	1.509**
	(0.792)	(0.252)	(0.810)	(0.652)	(0.222)	(0.665)
After $[2, 10]$	2.751	0.371	2.380	3.670	0.005	3.665
	(2.520)	(0.535)	(2.488)	(3.073)	(0.493)	(3.027)
After $[11, 20]$	4.710**	0.004	4.705**	3.837**	−0.029	3.866**
	(1.931)	(0.503)	(1.956)	(1.553)	(0.518)	(1.624)
After, full period $[2, 20]$	7.461*	0.396	7.065*	7.507*	−0.024	7.531*
	(4.182)	(0.843)	(4.120)	(4.319)	(0.863)	(4.284)
						Panel B: Departures
Before $[-20, -1]$	−1.403	−0.002	−1.401	−2.772	−0.109	−2.663
	(2.789)	(0.776)	(2.882)	(3.363)	(0.753)	(3.407)
Before $[-10, -1]$	−1.175	0.005	−1.180	−2.252	0.194	−2.446
	(1.690)	(0.537)	(1.805)	(2.111)	(0.505)	(2.205)
Before $[-5, -1]$	−2.712**	0.315	−3.027***	−2.724**	0.375	−3.099**
	(1.095)	(0.340)	(1.128)	(1.210)	(0.366)	(1.241)
Announcement $[0, +1]$	−1.150	0.050	−1.200	−1.388	−0.012	−1.375
	(0.839)	(0.183)	(0.829)	(0.874)	(0.186)	(0.853)
Announcement $[-1, +1]$	−1.861*	0.088	−1.949*	−1.575	0.067	−1.643
	(1.052)	(0.227)	(1.033)	(1.112)	(0.234)	(1.082)
After $[2, 10]$	0.434	−0.091	0.525	0.728	−0.463	1.191
	(2.535)	(0.663)	(2.625)	(1.918)	(0.530)	(2.029)
After $[11, 20]$	1.435	0.690	0.745	0.077	0.515	−0.438
	(2.086)	(0.735)	(2.087)	(2.027)	(0.701)	(2.053)
After, full period $[2, 20]$	1.869	0.599	1.270	0.806	0.052	0.754
	(3.489)	(1.166)	(3.661)	(3.217)	(1.029)	(3.368)
Observations (Arrivals)	81	867	948	81	858	939
Observations (Departures)	31	728	759	31	722	753

For arrivals, Panel A of Table A.3 shows positive and statistically significant same-industry returns under both expected-return models and announcement windows. Same-industry returns also exceed different-industry returns in each case. The two-day differences are 1.73 percentage points under the market model and 1.25 percentage points under the industry model, significant at the 1% and 5% levels, respectively. Thus, the announcement gains are concentrated among industry peers and do not depend on excluding the trading day immediately before the announcement.

Same-industry arrival returns are not statistically significant at conventional levels over $[-20,-1]$, $[-10,-1]$, or $[-5,-1]$ under either model in Panel A of Table A.3. The same-minus-different-industry returns are also not statistically significant at conventional levels, except for the industry-model difference of 1.72 percentage points over $[-5,-1]$, which is significant at the 10% level. These results do not rule out partial anticipation.

Stock return event studies measure how announcement news changes firm value (MacKinlay, 1997). A two-day window can therefore capture news about benefits expected to accrue over many years while limiting exposure to unrelated news. Anticipation may leave some of the relocation's value already reflected in prices, but the announcement return captures the additional revision (Malatesta and Thompson, 1985). The positive same-industry arrival returns and their significant difference from different-industry returns in Table A.3 show that favorable announcement news is concentrated among industry peers. If the announcement provides a common update about the probability of arrival and other news does not drive the differences, larger returns indicate greater expected benefits relative to firm value. The geographic concentration of the gains and their increase with product similarity in Section 4 further support the interpretation that investors expect local spillovers from the arriving headquarters.

Panel A of Table A.3 also shows appreciation after arrival announcements. Same-industry returns over $[2,10]$ are positive but statistically insignificant under both models. The gains over $[11,20]$ are 4.71% under the market model and 3.84% under the industry model, both significant at the 5% level. The same-minus-different-industry returns over this later subperiod are also signif-

icant at the 5% level. Over the full $[2,20]$ period, both the same-industry returns and the industry differences are positive and significant at the 10% level under both models. Different-industry incumbents show no comparable later appreciation. Thus, the concentration of gains among industry peers extends beyond the announcement, with stronger evidence in the later subperiod.

One possible explanation for the later arrival gains is that investors take time to assess how a headquarters move affects nearby firms. DellaVigna and Pollet (2009) and Hirshleifer et al. (2009) document earnings-announcement responses followed by returns in the same direction.[17] Cohen and Frazzini (2008) show that information about economically connected firms can enter prices with a delay. Subsequent reporting may also convey new information. The return paths cannot distinguish these explanations from unrelated news arriving over the longer window. The positive two-day response measures the revision around the announcement even if investors continue to assess the arrival afterward. The upward revision at the announcement therefore does not require attributing the later appreciation to the relocation.

For departures, Panel B of Table A.3 shows same-industry losses of 2.71% under the market model and 2.72% under the industry model over $[-5,-1]$, both significant at the 5% level. The corresponding industry differences are negative and significant under both models. These losses are consistent with some adjustment before the announcement. The two-day announcement returns and industry differences are negative but statistically insignificant. Including the preceding trading day makes the market-model return and industry difference significant at the 10% level. The corresponding industry-model estimates are not statistically significant at conventional levels.

After departure announcements, same-industry returns and industry differences are statistically insignificant in every reported window. Panel B of Table A.3 therefore provides no clear evidence of continued losses or a reversal. Thus, departure losses are clearest before the announcement, while arrivals provide stronger evidence of an additional valuation revision at the announcement.

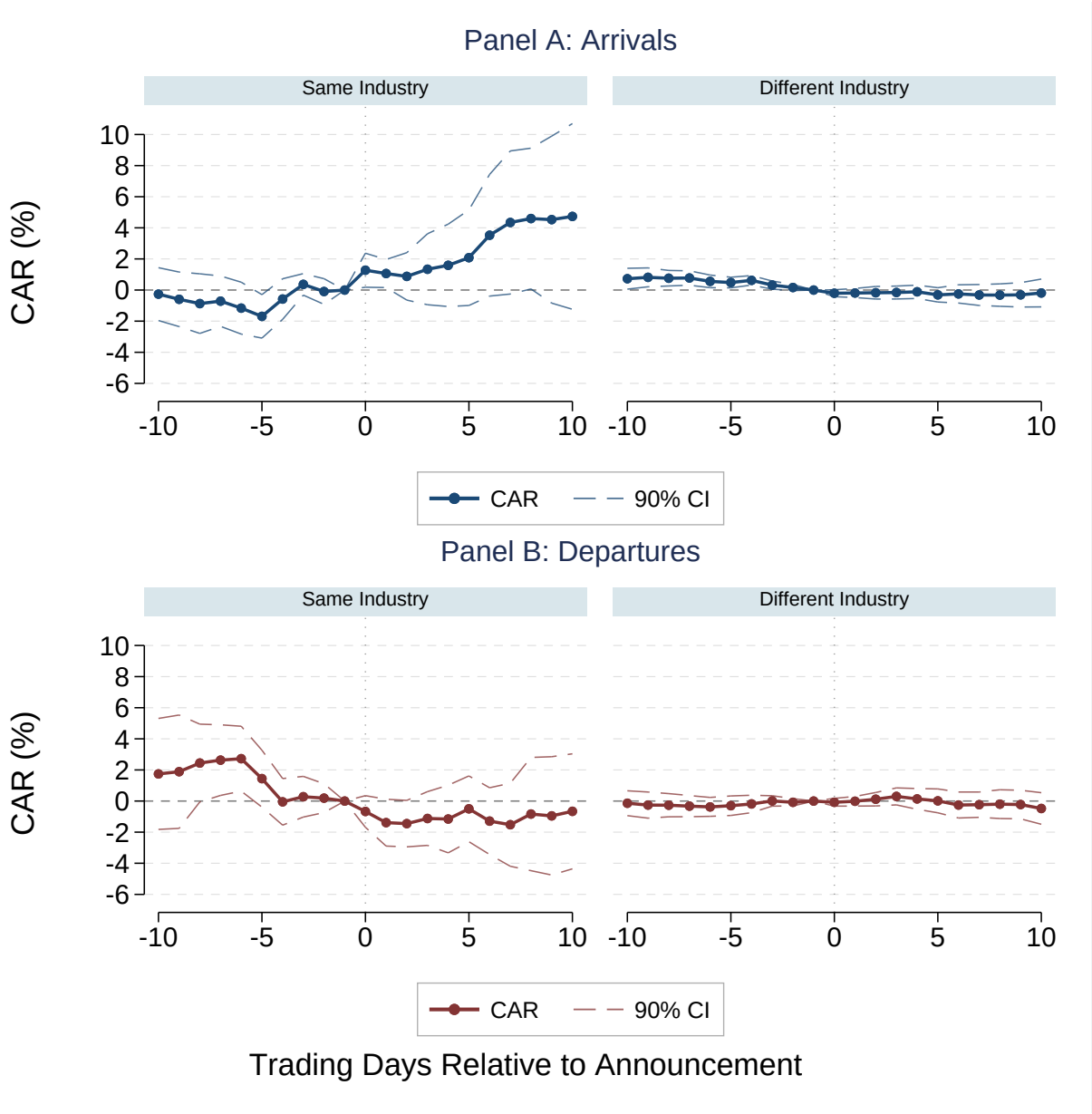
[17] Hirshleifer, Lim, and Teoh distinguish $[0,+1]$ and $[2,61]$ returns. DellaVigna and Pollet distinguish $[0,+1]$ and $[2,75]$ returns. Both studies use compounded returns. The present analysis retains the arithmetic CAR measure used in the main paper.

The difference in statistical significance does not by itself establish that the arrival and departure effects differ.

The paths in Figure 5 are normalized to zero on day -1, so the value on day 1 equals the two-day announcement return. This normalization does not imply that values were unchanged before the announcement. Pointwise confidence intervals describe uncertainty in the estimated mean cumulative return at each event day, while Table A.3 reports the window and industry-difference tests. Longer windows generally admit more return variation and unrelated news (Kothari and Warner, 2007). The short window therefore captures a valuation update around the announcement while limiting exposure to other news.

Figure A.2 reports the corresponding daily paths using the Fama–French 48-industry model. Same-industry arrival returns remain positive after the announcement and generally rise further, while different-industry incumbents show no comparable appreciation. Same-industry departure returns remain negative after the announcement but are generally not statistically significant at conventional levels. The industry adjustment therefore preserves the qualitative timing patterns in Figure 5, suggesting that national industry returns do not fully account for the arrival gains. Table A.3 reports the formal comparisons between industry groups.

These timing tests show that same-industry incumbents gain value around arrival announcements and significantly outperform incumbents in other industries. Interpreting this pattern as news about the arrival requires a reasonable expected-return benchmark and that unrelated incumbent news not systematically coincide with the announcement, as discussed in Section 3. Under these assumptions, the returns indicate that investors expect nearby industry peers to benefit from the arrival. The conclusion allows for both partial anticipation and continued adjustment after the announcement. For corporate-attraction policy, the returns therefore help identify the types of existing firms expected to benefit. Differences in anticipation or adjustment across firms limit comparisons of total benefits, so the two-day response need not rank firms by the full value they gain from the relocation.



Appendix Figure A.2. Industry-Adjusted Incumbent Return Paths Around Headquarters Relocation Announcements

Notes: This figure shows mean cumulative abnormal returns, in percent, from the Fama–French 48-industry model for public incumbents from ten trading days before through ten trading days after headquarters relocation announcements. Each path is normalized to zero on day −1. Panel A shows destination-side incumbents around arrivals. Panel B shows origin-side incumbents around departures. Same-industry incumbents on the left side of each panel share the relocating headquarters' ZIP code and four-digit NAICS industry. Different-industry incumbents on the right side of each panel share the ZIP code but operate in other four-digit industries. The announcement-window samples contain 81 same-industry and 858 different-industry arrival firm-events, and 31 same-industry and 722 different-industry departure firm-events. Daily paths use available firm-event returns. The vertical scale matches the market-model figure in the main paper. Solid lines show mean cumulative abnormal returns. Dashed lines trace pointwise 90% confidence bounds using standard errors clustered by relocation announcement and t critical values.

B.3 Returns on Placebo and Future Comparison Dates

I next compare the announcement response with returns for the same incumbents on dates away from the announcement. Table A.4 reports the comparison-date returns separately for arrivals in Panel A and departures in Panel B. Column [1] reports average returns on random comparison dates. Columns [2]–[4] report returns one year, eighteen months, and two years after the announcement. All entries use a two-day $[0, +1]$ window relative to the comparison date.

For both arrivals and departures, eligible random dates lie 141 to 252 trading days before or after the actual announcement. The use of hypothetical announcement dates is motivated by Brown and Warner (1985). I estimate the market and industry models separately before each comparison date using its preceding $[-120, -31]$ interval. A date is eligible only if every incumbent associated with the relocation has sufficient data.[18] I average each incumbent's two-day abnormal return across all eligible dates, giving the mean return that uniform random selection would produce. This comparison retains all 81 arrival and 31 departure observations.

For the future-date comparisons in Table A.4, I use the first trading day on or after 12, 18, and 24 calendar months from the announcement and re-estimate expected returns before each date. Sample sizes vary with the availability of later returns and are reported in the table. Within each comparison, both return models use the same firms and the original weights. These comparisons assess returns at other dates and do not measure whether the initial gain persists between those dates.

[18] I require complete $[-1, +1]$ returns, both return benchmarks, and at least 80 valid observations in each model's estimation interval. The comparison estimation interval excludes the actual $[-20, 20]$ period, and the comparison return window excludes the original estimation interval. Each relocation has 112 to 224 eligible dates, each assigned equal probability. All incumbents associated with a relocation use the same comparison date. Available histories determine the balance of eligible dates before and after the announcement.

Appendix Table A.4. Incumbent Returns on Placebo and Future Comparison Dates

Notes: This table reports mean two-day $[0, +1]$ cumulative abnormal returns (in percent) for same-ZIP, four-digit NAICS incumbents on placebo and future comparison dates. Column [1] averages returns across eligible dates 141–252 trading days before or after the announcement. Columns [2]–[4] use the first trading day on or after 12, 18, and 24 calendar months after the announcement. The market and Fama–French 48-industry models are reestimated before each date and use the same observations within each column. Standard errors clustered by relocation announcement appear in parentheses. Significance at 1%, 5%, and 10% is denoted by ***, **, and * respectively.

	Random Dates	+1 Year	+18 Months	+2 Years
	[1]	[2]	[3]	[4]
				Panel A: Arrivals
Market Model	0.108	−0.007	0.067	0.167
	(0.072)	(0.429)	(0.907)	(0.354)
Industry Model	0.038	−0.929**	−0.458	0.144
	(0.058)	(0.354)	(0.532)	(0.299)
				Panel B: Departures
Market Model	−0.019	−0.302	−0.099	−0.260
	(0.259)	(1.001)	(0.995)	(1.260)
Industry Model	−0.109	−0.314	−0.349	0.244
	(0.271)	(1.017)	(0.796)	(1.034)
Observations (Arrivals)	81	81	78	78
Observations (Departures)	31	30	29	25

Random-date mean returns in Table A.4 are close to zero and statistically insignificant for both arrivals and departures under both models. Returns on the selected future dates vary across horizons and are generally not statistically significant at conventional levels.

Table A.5 compares announcement returns with these benchmarks for arrivals in Panel A and departures in Panel B. Within each model, the rows report the announcement return, the comparison-date return, and their difference. Each column uses the same incumbent–announcement observations and weights for all three rows. The differences therefore compare returns for the same firms at the announcement and on the comparison dates.

Appendix Table A.5. Two-Day Announcement Returns Relative to Comparison Dates

Notes: This table compares two-day $[0, +1]$ announcement returns with returns on the dates defined in Table A.4. Within each column, both models use the same incumbents for announcement returns, comparison returns, and their difference. Returns are in percent and differences in percentage points. The random-date benchmark is each incumbent's mean return across eligible dates. Standard errors clustered by relocation announcement appear in parentheses. Significance at 1%, 5%, and 10% is denoted by ***, **, and *, respectively.

	Random Dates	+1 Year	+18 Months	+2 Years
	[1]	[2]	[3]	[4]
				Panel A: Arrivals
Market Model				
Actual announcement	1.401**	1.401**	1.455**	1.455**
	(0.534)	(0.534)	(0.546)	(0.546)
Comparison return	0.108	−0.007	0.067	0.167
	(0.072)	(0.429)	(0.907)	(0.354)
Actual minus comparison	1.293**	1.408**	1.388	1.288**
	(0.528)	(0.635)	(1.137)	(0.523)
Industry Model				
Actual announcement	1.063*	1.063*	1.132**	1.132**
	(0.526)	(0.526)	(0.514)	(0.514)
Comparison return	0.038	−0.929**	−0.458	0.144
	(0.058)	(0.354)	(0.532)	(0.299)
Actual minus comparison	1.025*	1.992**	1.590**	0.988*
	(0.546)	(0.785)	(0.684)	(0.540)
				Panel B: Departures
Market Model				
Actual announcement	−1.150	−1.133	−1.162	−1.482
	(0.839)	(0.866)	(0.896)	(1.008)
Comparison return	−0.019	−0.302	−0.099	−0.260
	(0.259)	(1.001)	(0.995)	(1.260)
Actual minus comparison	−1.131	−0.831	−1.063	−1.221
	(0.958)	(1.528)	(0.924)	(1.639)
Industry Model				
Actual announcement	−1.388	−1.366	−1.374	−1.771
	(0.874)	(0.901)	(0.933)	(1.046)
Comparison return	−0.109	−0.314	−0.349	0.244
	(0.271)	(1.017)	(0.796)	(1.034)
Actual minus comparison	−1.279	−1.053	−1.025	−2.015
	(1.010)	(1.664)	(0.821)	(1.463)
Observations (Arrivals)	81	81	78	78
Observations (Departures)	31	30	29	25

For arrivals, Panel A of Table A.5 shows that announcement returns exceed the random-date averages by 1.29 percentage points under the market model and 1.03 percentage points under the industry model, significant at the 5% and 10% levels, respectively. Announcement returns also exceed returns one year later under both models, with both differences significant at the 5% level. The differences remain positive at the later comparison dates, although their precision varies. Thus, the arrival gains exceed the same incumbents' mean abnormal returns on eligible random dates and their returns on selected future dates.[19]

For departures, Panel B of Table A.5 shows announcement returns below the random-date averages and below returns at each future horizon under both models. All differences are statistically insignificant. The comparison-date evidence is therefore stronger for arrivals.

These comparisons strengthen the evidence that arrival announcements convey favorable news for nearby industry peers. The announcement gains exceed the same incumbents' mean abnormal returns on eligible random dates, while Table A.3 shows that their announcement returns significantly exceed those of firms in other industries. Under the identification assumptions in Section 3, these results support an upward revision in the value of local spillovers when the destination gains an industry peer.

[19]The random-date tests compare announcement returns with the mean across eligible dates. In a separate exercise, I draw one eligible date per relocation in each of 1,000 assignments. The actual two-day arrival and departure estimates lie within the central 95% of these distributions. A significant difference from the mean benchmark therefore does not establish that the announcement estimate lies in the tail of the distribution across individual date assignments.

B.4 Announcement Timing and Contemporaneous News

Table A.6 examines contemporaneous incumbent news by excluding firm-events with an 8-K filing within two calendar days of the relocation announcement. This restriction removes observations with nearby incumbent filings that could contribute to the measured return. Arrival returns remain positive, but the two-day estimates fall to 0.83% under the market model and 0.56% under the industry model and become statistically insignificant. The restricted arrival sample contains 62 observations, compared with 81 in the full sample. Departure returns remain negative under both models. Thus, the restriction preserves the direction of the announcement responses but weakens the arrival evidence.

Appendix Table A.6. Incumbent Returns Excluding Contemporaneous Filings

Notes: This table reports cumulative abnormal returns for public incumbents in the relocating firm's ZIP code and four-digit NAICS industry, excluding firms with Form 8-K filings within two calendar days of the announcement. Columns [1] and [3] use the market model and [2] and [4] the Fama–French 48-industry model (Fama and French, 1997). Standard errors clustered by relocation announcement appear in parentheses. Significance at 1%, 5%, and 10% is denoted by ***, **, and * respectively.

	$[0,+1]$		$[-1,+1]$	
	[1]	[2]	[3]	[4]
Panel A: Arrivals				
CAR, %	0.83	0.56	0.97	0.83
	(0.52)	(0.48)	(0.88)	(0.73)
Panel B: Departures				
CAR, %	-1.44	-1.73*	-2.40**	-2.05*
	(0.91)	(0.84)	(1.06)	(1.05)
Obs. (Arrivals)	62	62	62	62
Obs. (Departures)	25	25	25	25
Fama–French 48-industry model	No	Yes	No	Yes

B.5 Leave-One-Out Estimates

Table A.7 assesses the influence of individual announcements by re-estimating market-model returns after omitting all incumbents associated with one relocation at a time. This test asks whether the direction of the announcement response depends on a single move.

Every omission preserves the positive arrival and negative departure estimates across the reported windows. Two-day arrival returns range from 0.88% to 1.53%, while departure returns range from -1.44% to -0.51%. Thus, no single relocation determines the direction of the estimates. These ranges describe the estimated returns and do not establish statistical significance in every restricted sample.

Appendix Table A.7. Leave-One-Out Sensitivity of Incumbent Returns

Notes: This table reports market-model cumulative abnormal returns for public incumbents in the relocating firm's ZIP code and four-digit NAICS industry. The leave-one-out range spans estimates obtained by omitting one relocation announcement at a time. Standard errors clustered by relocation announcement appear below the full-sample estimates in parentheses. Significance at 1%, 5%, and 10% is denoted by ***, **, and * respectively.

	Arrivals		Departures	
	$[0,+1]$	$[-1,+1]$	$[0,+1]$	$[-1,+1]$
	[1]	[2]	[3]	[4]
Full-sample CAR, %	1.40**	1.36*	-1.15	-1.86*
	(0.53)	(0.79)	(0.84)	(1.05)
Leave-one-out range, %	[0.88, 1.53]	[0.81, 1.55]	[-1.44, -0.51]	[-2.37, -1.34]
Obs.	81	81	31	31

C Patenting Outcomes

I examine whether the patenting results in Section 5 are sensitive to including the relocation year and to measuring the baseline before the public announcement. These timing comparisons help assess the innovation gains associated with attracting a headquarters.

C.1 Timing of Patenting Outcomes

To test if the main findings are driven by the exclusion of the transition year, I re-estimate the patenting outcomes with the year of the relocation included in the post-treatment period. Table A.8 re-estimates the two specifications from Table 3, adding the first destination 10-K year into the post-treatment window. Because this year spans the headquarters transition, these estimates test whether the positive patenting pattern depends on omitting the actual move year. In this test, I show that including the transition year attenuates the estimates, which is expected given that newly arrived headquarters have limited time to influence incumbent innovation during the physical move. Nevertheless, the cohort fixed-effects Poisson specification shows a statistically significant increase of 20 patents in the three-mile sample and 22 patents in the 10-mile sample. The linear estimates are 11 and 12 patents in column [2]. Both linear estimates remain statistically insignificant. The positive patenting response therefore extends to the window that includes the move year, with greater precision under the Poisson specification.

Appendix Table A.8. Incumbent Patenting Including the Transition Year

Notes: This table shows average changes in annual patent production following headquarters arrivals over 2004–2024. Exposed firms share the arriving headquarters' ZIP code and four-digit NAICS industry. Comparison firms are never-treated peers in destination MSAs located at least 3 miles (Panel A) or 10 miles (Panel B) away. Column [1] reports Poisson estimates with cohort and year fixed effects (Wooldridge, 2023). Column [2] reports additive group-time estimates using outcome regression (Callaway and Sant'Anna, 2021), which difference out time-invariant unobserved heterogeneity. All columns allow cohort-specific treatment effects and include the relocation year. Estimates average effects from the relocation year through 3 years after relocation. Standard errors clustered by incumbent ZIP code are in parentheses. Significance at 1%, 5%, and 10% is denoted by ***, **, and *, respectively.

	Poisson	Callaway–Sant'Anna
	[1]	[2]
Panel A: Three-mile comparison group		
Change in annual patents	20.2**	11.0
	(8.4)	(13.4)
Obs.	5,211	5,347
Panel B: Ten-mile comparison group		
Change in annual patents	22.2**	12.0
	(9.4)	(13.3)
Obs.	3,354	3,490
Year FE	Yes	No
Cohort FE	Yes	No

I also examine relocations whose announcement and first destination 10-K observation occur in the same calendar year, so that patenting in the preceding year is measured before the public announcement. This restriction retains 62 of the 94 eligible treated patenting firms.

Table A.9 reports the results. The cohort Poisson estimates imply 39 and 43 additional annual patents under the three-mile and ten-mile restrictions, respectively (column [1]). Both Poisson estimates are significant at 1%. The linear estimates are 24 and 26 patents in column [2]. Both linear estimates remain statistically insignificant.[20] The positive Poisson response therefore also

[20] Allowing for MSA-specific comparison changes gives estimates of 11 and 13 patents when the transition year is included, and 24 and 25 patents for same-year relocations, under the three-mile and ten-mile restrictions, respectively, and all four estimates remain statistically insignificant.

appears with a baseline that precedes the announcement. For evaluating the innovation associated with these arrivals, the estimates describe a smaller set of incumbent firms exposed to this subset of same-year relocations.

Appendix Table A.9. Incumbent Patenting for Same-Year Relocations

Notes: This table shows average changes in annual patent production following headquarters arrivals over 2004–2024. Exposed firms share the arriving headquarters' ZIP code and four-digit NAICS industry. Comparison firms are never-treated peers in destination MSAs located at least 3 miles (Panel A) or 10 miles (Panel B) away. Only relocations announced and first observed in a destination 10-K in the same calendar year are included, so the baseline precedes the public announcement. Column [1] reports Poisson estimates with cohort and year fixed effects (Wooldridge, 2023). Column [2] reports additive group-time estimates using outcome regression (Callaway and Sant'Anna, 2021), which difference out time-invariant unobserved heterogeneity. All columns allow cohort-specific treatment effects and exclude the relocation year. Estimates average effects from 1 to 3 years after relocation. Standard errors clustered by incumbent ZIP code are in parentheses. Significance at 1%, 5%, and 10% is denoted by ***, **, and *, respectively.

	Poisson	Callaway–Sant'Anna
	[1]	[2]
Panel A: Three-mile comparison group		
Change in annual patents	38.9***	23.5
	(4.5)	(28.2)
Obs.	4,996	5,090
Panel B: Ten-mile comparison group		
Change in annual patents	42.9***	25.5
	(2.6)	(28.5)
Obs.	3,139	3,233
Year FE	Yes	No
Cohort FE	Yes	No